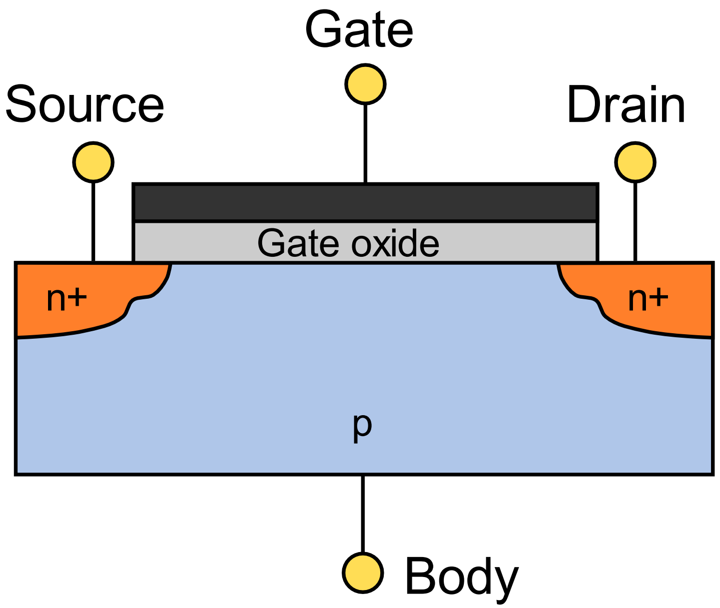
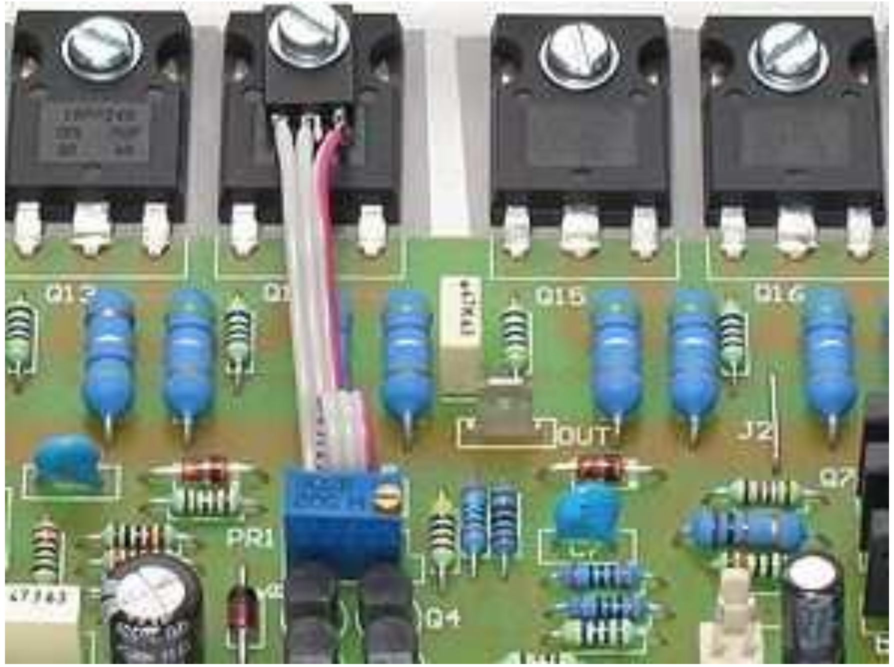




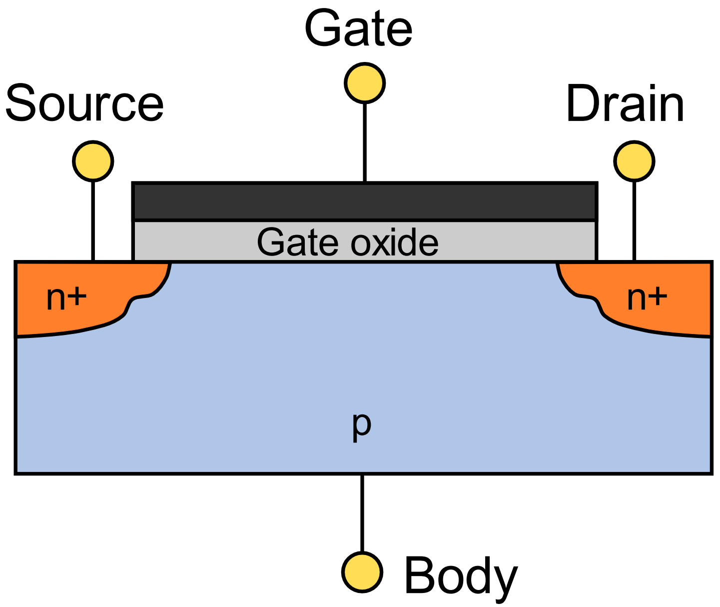
Field Effect Transistor (FET)

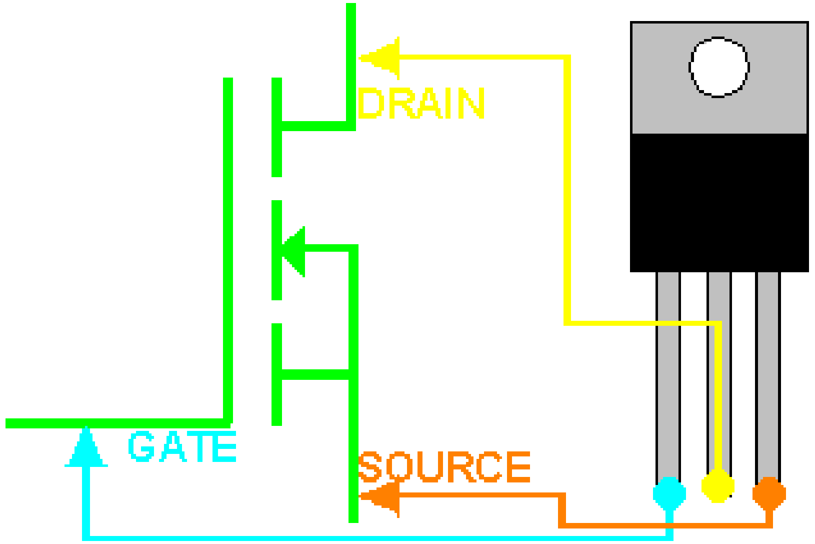
Field Effect Transistor (FET)

A **Field Effect Transistor (FET)** is a **voltage-controlled semiconductor device** that controls the flow of current using an electric field. Unlike bipolar junction transistors (BJTs), which are current-controlled, FETs use voltage to control the current flowing through a channel.

A FET has **three main terminals**:

- **Source (S):** Terminal through which carriers enter the channel.
- **Drain (D):** Terminal through which carriers leave the channel.
- **Gate (G):** Controls the width of the channel and hence the current flow by applying voltage.





Field Effect Transistor (FET) Cont.

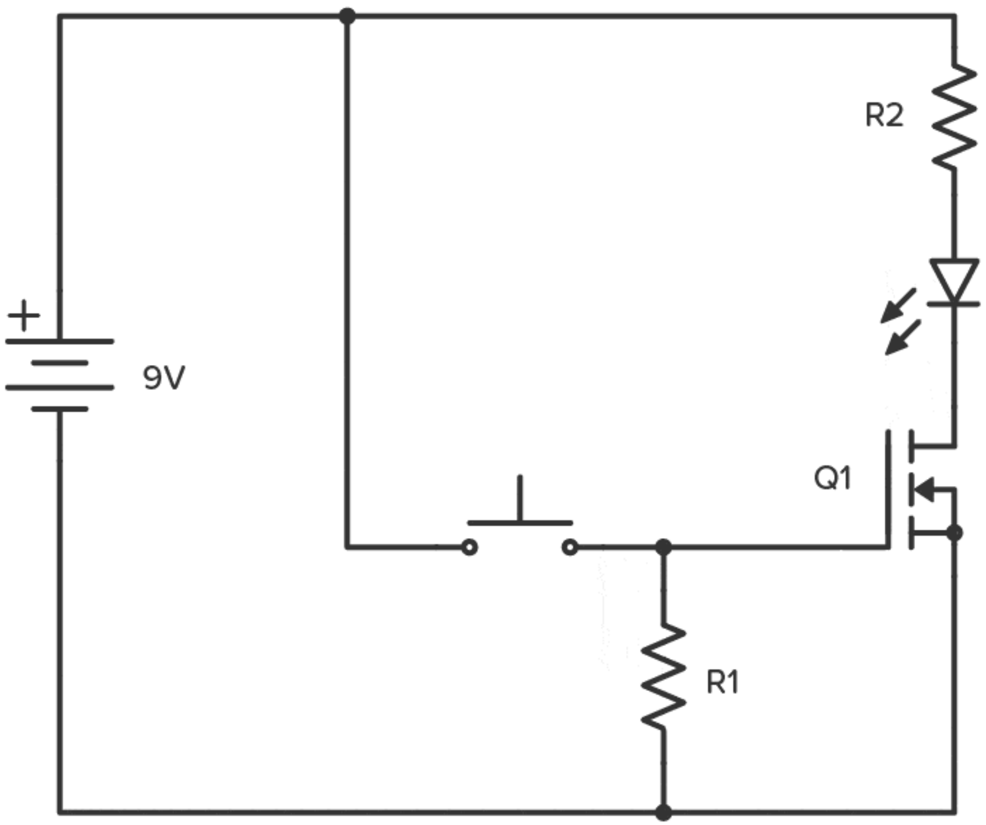
There are two major types of channels:

- **n-channel:** Uses electrons as charge carriers.
- **p-channel:** Uses holes as charge carriers.

Operating Principle

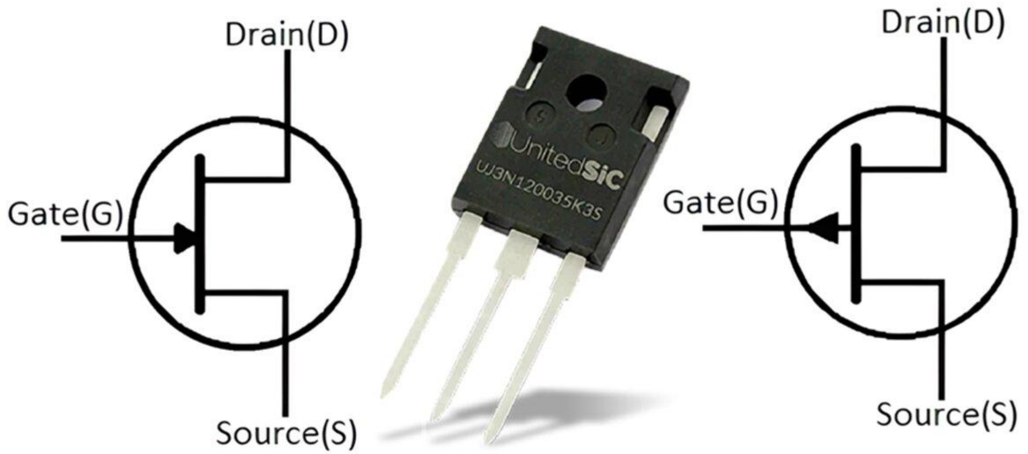
The operation of a FET is based on the **control of channel conductivity** between the **source and drain** by an electric field induced by a voltage at the **gate terminal**.

- When a voltage is applied to the gate, it creates an electric field that **either attracts or repels carriers** in the channel.
- This modulates the **resistance of the channel**, allowing more or less current to flow between source and drain.

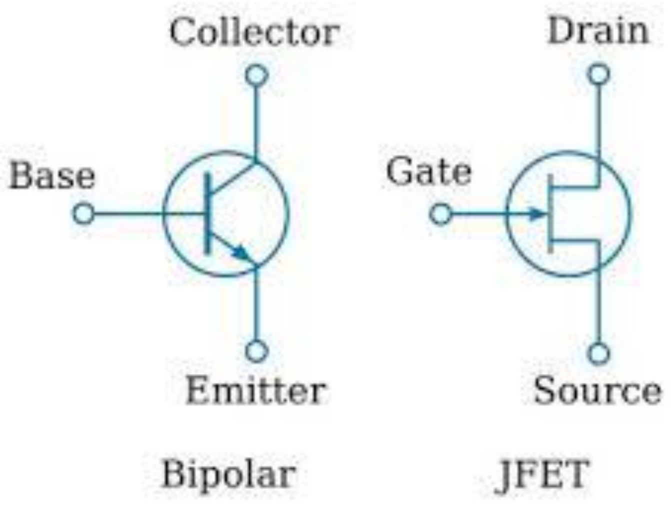


Types of FETs: Junction Field Effect Transistor (JFET)

- **Normally ON** (depletion-type)
- Gate is **reverse biased** to control current.
- Two types:
  - **n-channel JFET**
  - **p-channel JFET**
- Has a **pn-junction** between the gate and the channel.

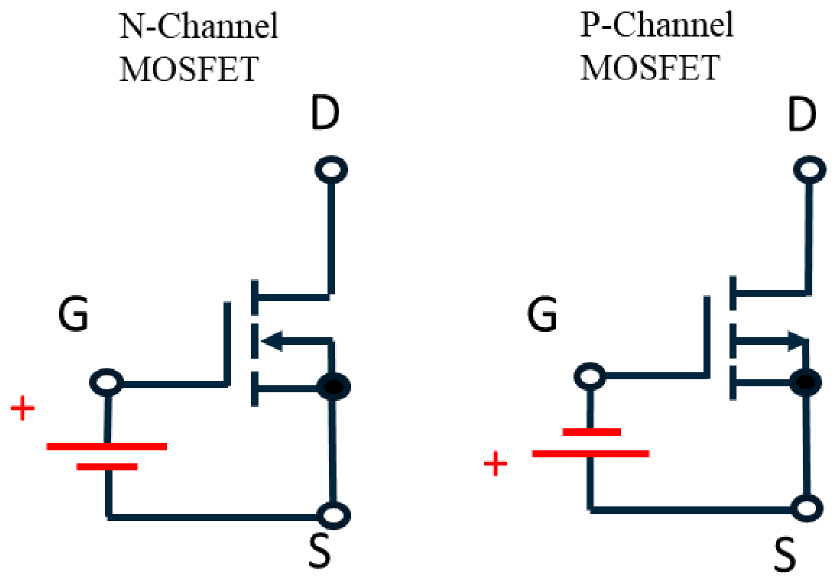


N-channel JFET P-channel JFET



Types of FETs: Metal Oxide Semiconductor Field Effect Transistor (MOSFET)

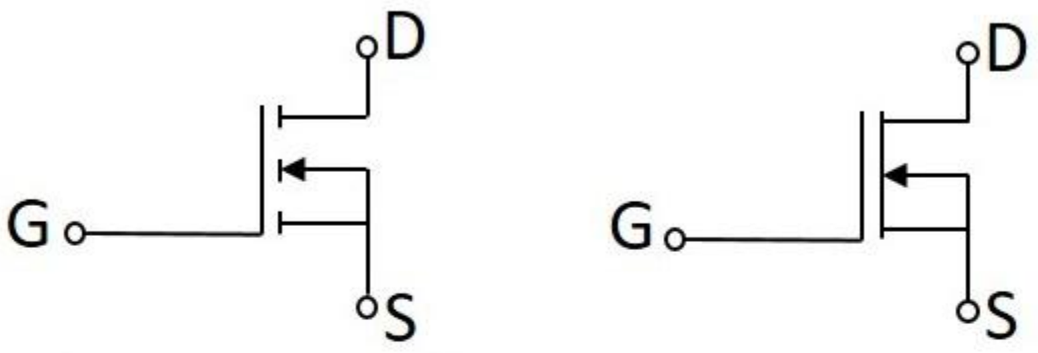
- Most widely used FET.
- Can be **enhancement-mode** (normally OFF) or **depletion-mode** (normally ON).
- Has an **insulated gate** (typically with a thin oxide layer).
- Two main types:
  - **n-channel MOSFET (NMOS)**
  - **p-channel MOSFET (PMOS)**



Enhancement vs Depletion MOSFETs

Mode	Channel Exists at Zero Gate Voltage?	Normally ON or OFF?	Gate Bias for Conduction
Enhancement Mode	No	OFF	Positive (NMOS), Negative (PMOS)
Depletion Mode	Yes	ON	Negative (NMOS), Positive (PMOS)

Symbols of N-Channel MOSFET

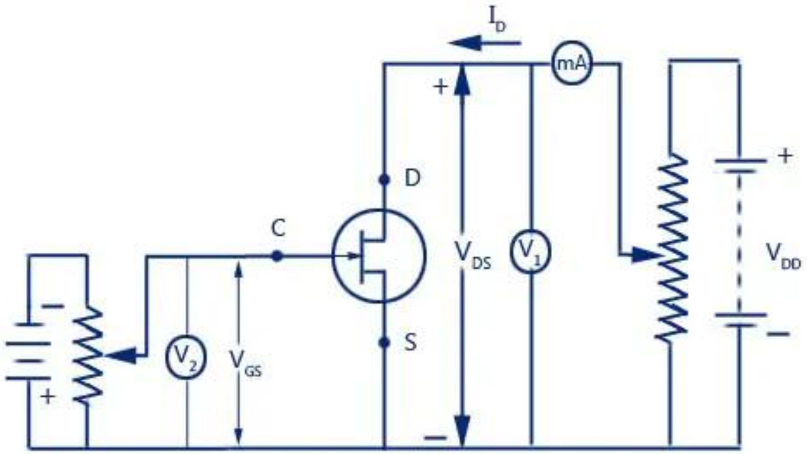


Enhancement Mode Depletion Mode

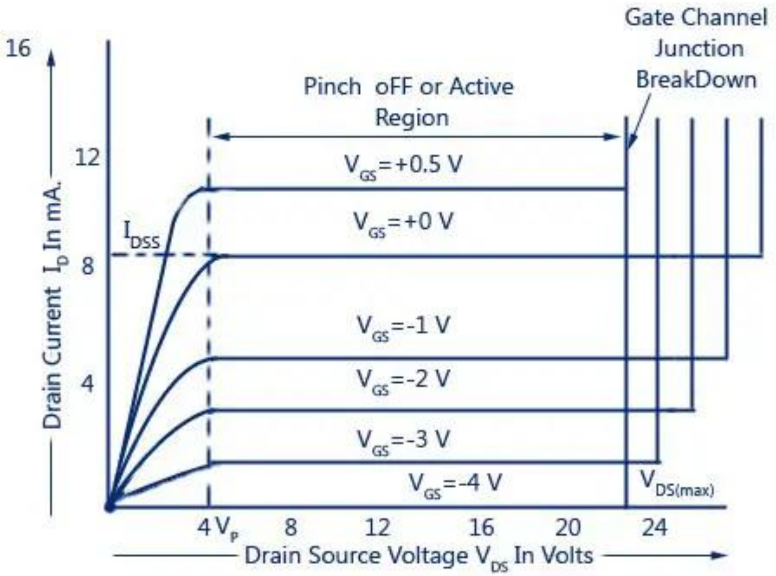
VI Characteristics

Output Characteristics (I_D vs V_{DS}): Shows the drain current vs drain-source voltage at different gate voltages.

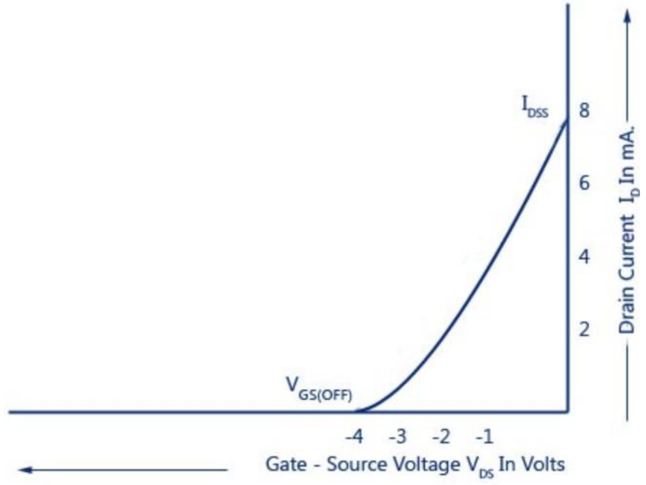
Transfer Characteristics (I_D vs V_{GS}): Shows how drain current varies with gate-source voltage.



Circuit Diagram for Determining Drain Characteristic with External-Bias for An N-Channel JFET



Drain Characteristic with External-Bias



Transfer Characteristics of JFET

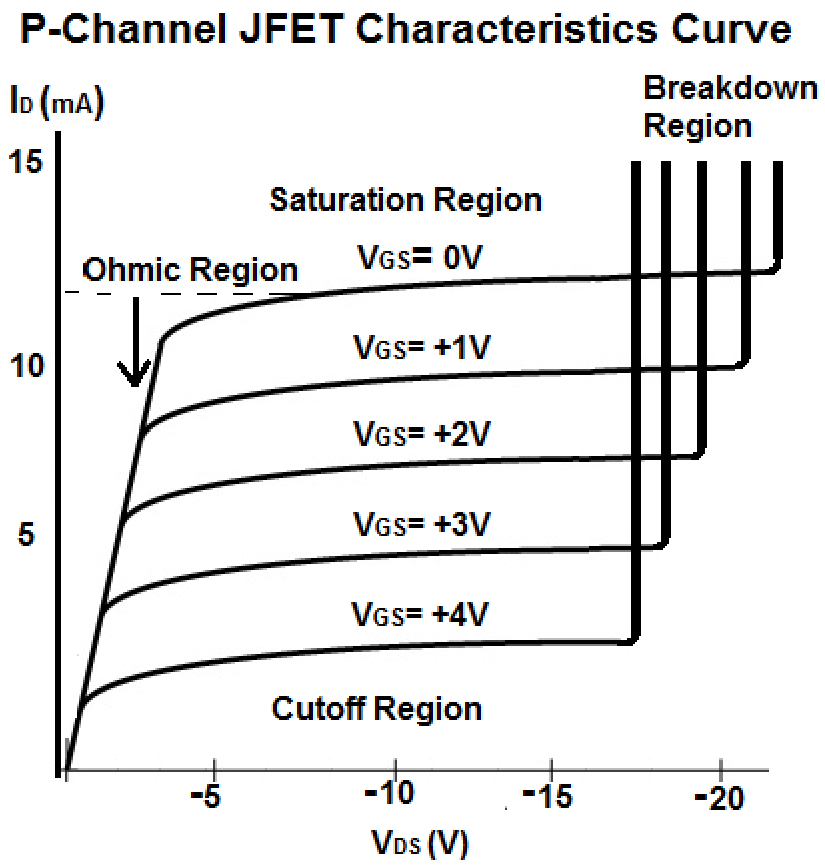
External Bias Characteristic of JFET

VI Characteristics

Output Characteristics (I_D vs V_{DS}): Shows the drain current vs drain-source voltage at different gate voltages.

FETs exhibit three regions:

- **Ohmic (Linear) Region:** FET behaves like a variable resistor.
- **Saturation (Active) Region:** Current remains constant; used for amplification.
- **Cut-off Region:** No current flows; device is OFF.



Advantages of FETs

- High input impedance (almost infinite for MOSFETs).
- Low power consumption.
- Less thermal noise.
- Better for integration in ICs.
- Voltage-controlled operation makes it simpler for digital circuits.

Disadvantages of FETs

- Susceptibility to damage from static discharge (especially MOSFETs).
- Lower gain compared to BJTs in some cases.
- Higher on-resistance in some applications.

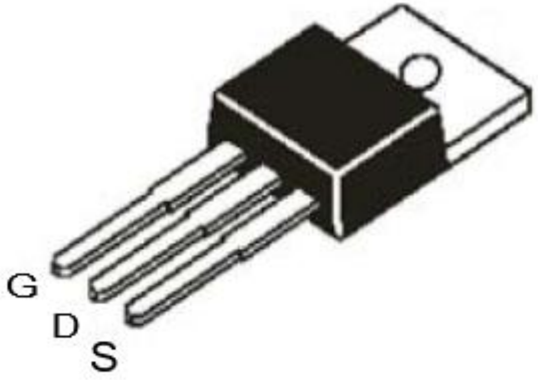
Applications of FETs

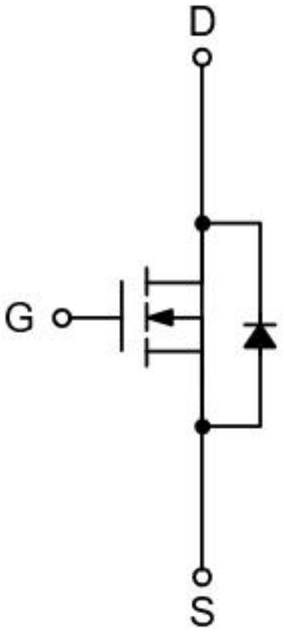
- **Amplifiers**: In RF, audio, and instrumentation.
- **Switching circuits**: In digital electronics, especially MOSFETs.
- **Analog switches**: In signal routing.
- **Voltage-controlled resistors**: In analog control circuits.
- **Power electronics**: Power MOSFETs are used in high-current, high-speed switching applications.

Power FETs and Variants

- **Power MOSFETs**: Used in power supplies, motor control.
- **DMOS (Double-diffused MOSFET)**: Better performance at high voltages.
- **IGBT (Insulated Gate Bipolar Transistor)**: Combines the ease of MOSFET control with BJT current handling.
- **MESFET (Metal Semiconductor FET)**: Used in microwave and RF applications.
- **FinFETs**: 3D transistors used in modern nano-scale integrated circuits.

Example: HY1707P - N-Channel Enhancement Mode MOSFET





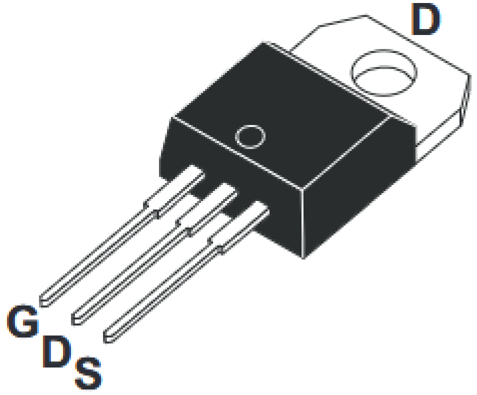
70V/80A,

$R_{DS(ON)}$ = 6mΩ (typ.) @ V_{GS}=10V

Power Management for Inverter Systems.

N-Channel MOSFET

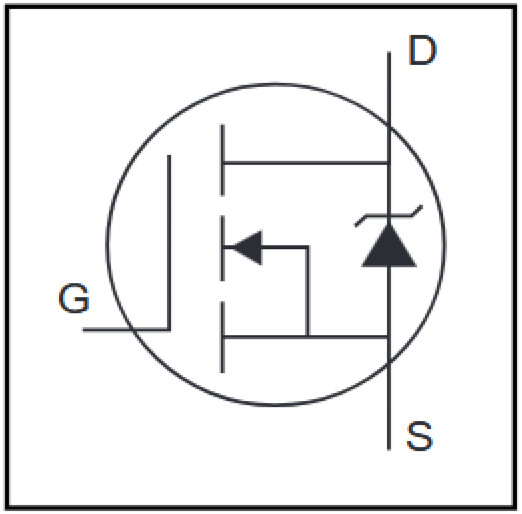
IRF3205 N-Channel Power MOSFET



The Nell **IRF3205** is a three-terminal silicon device with current conduction capability of 110A, fast switching speed, low on-state resistance, breakdown voltage rating of 55V, and max. threshold voltage of 4 volts.

They are designed as an extremely efficient and reliable device for use in a wide variety of applications. These transistors can be operated directly from integrated circuits.

HEXFET® Power MOSFET

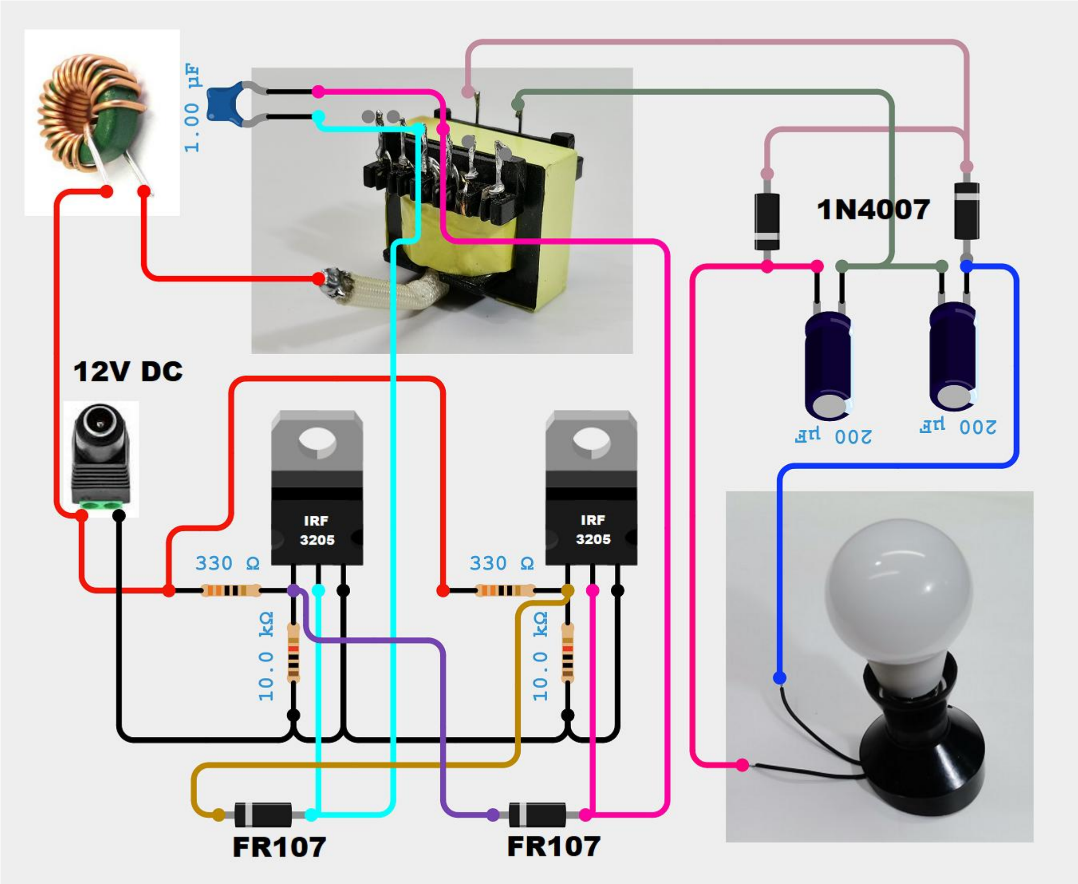


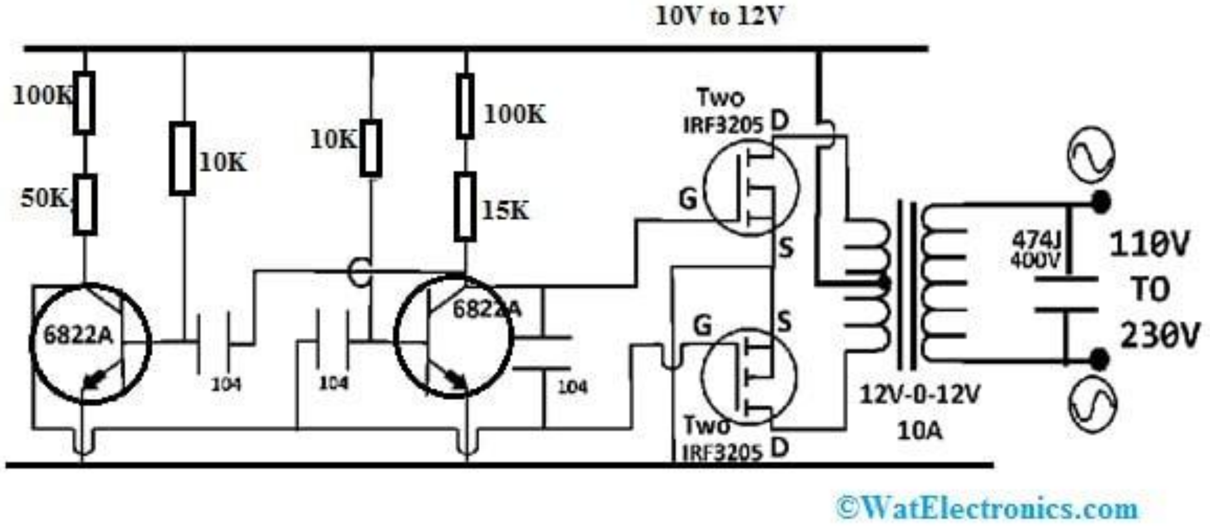
$V_{DSS} = 55V$

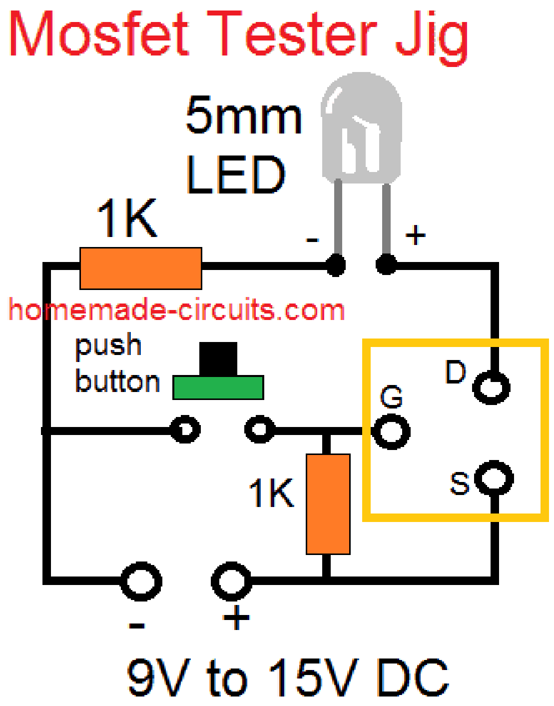
$R_{DS(on)} = 8.0m\Omega$

$I_D = 110A^{⑤}$

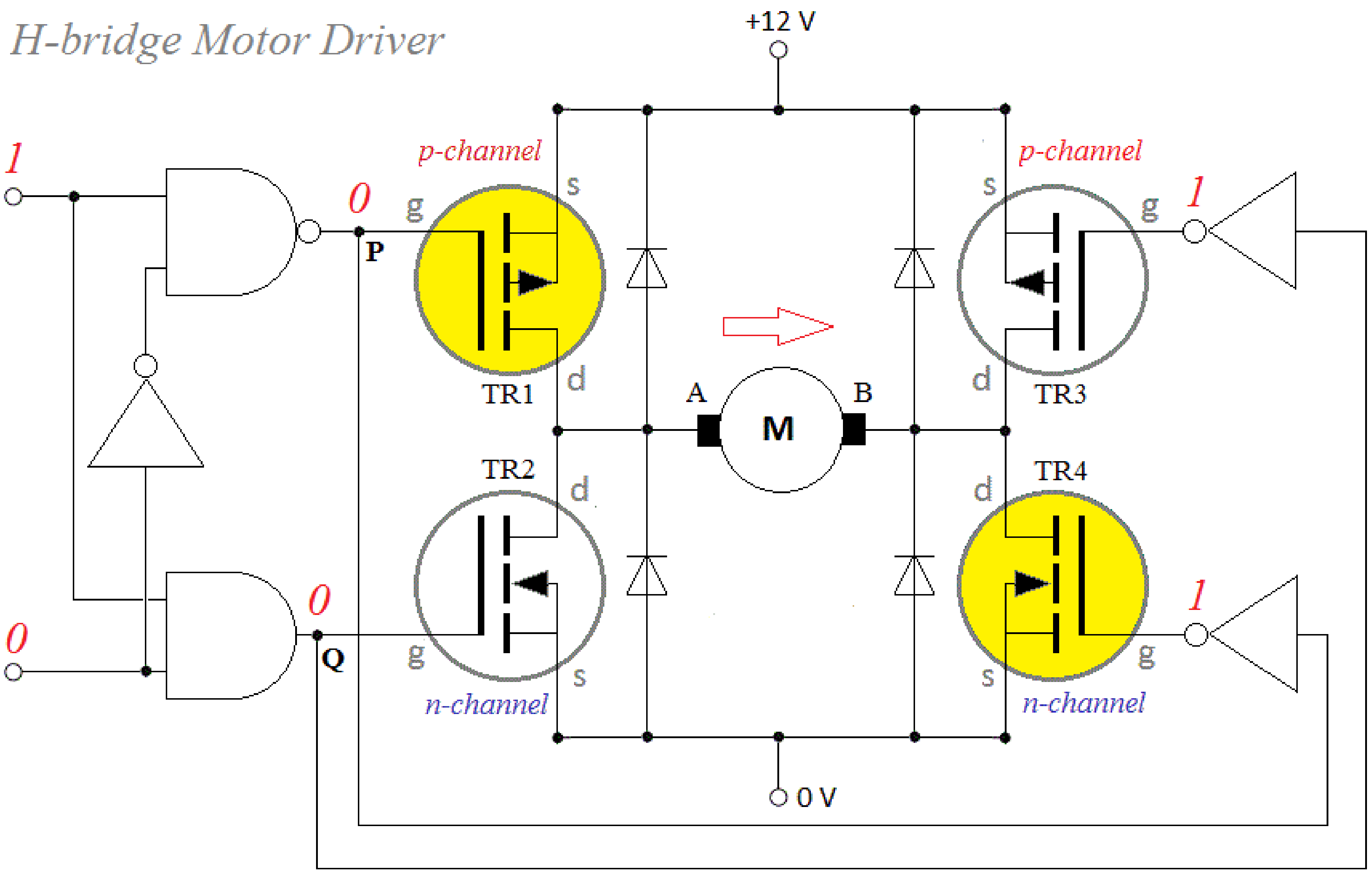
IRF3205 N-Channel Power MOSFET Cont.







H-bridge Motor Driver



Comparison

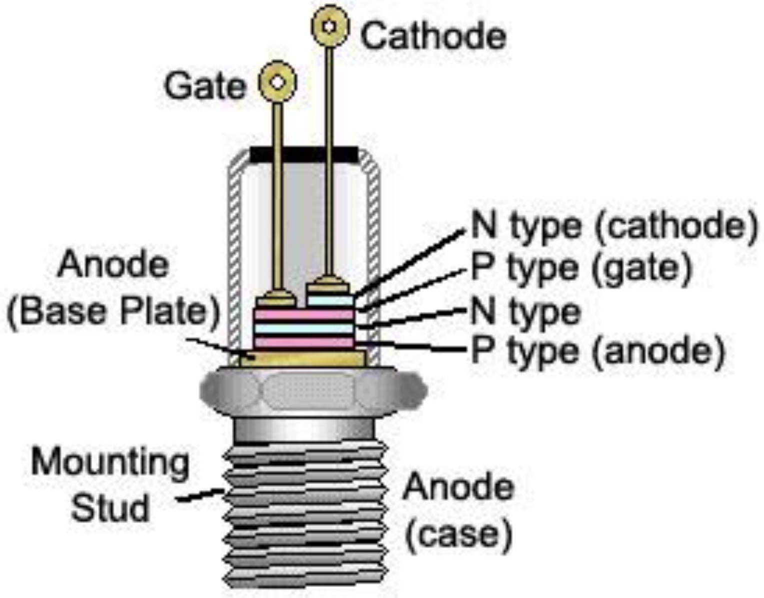
Feature	BJT	JFET	MOSFET
Type	Bipolar	Unipolar	Unipolar
Control	Current	Voltage	Voltage
Input Impedance	Low	High	Very High
Switching Speed	Moderate	Moderate	Very High
Noise	Higher	Lower	Lowest
Power Handling	Good	Moderate	Excellent (esp. Power MOSFETs)
Integration in ICs	Difficult	Moderate	Excellent
Cost	Generally low	Low to medium	Low to high

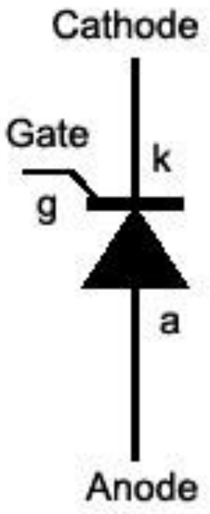
Outline

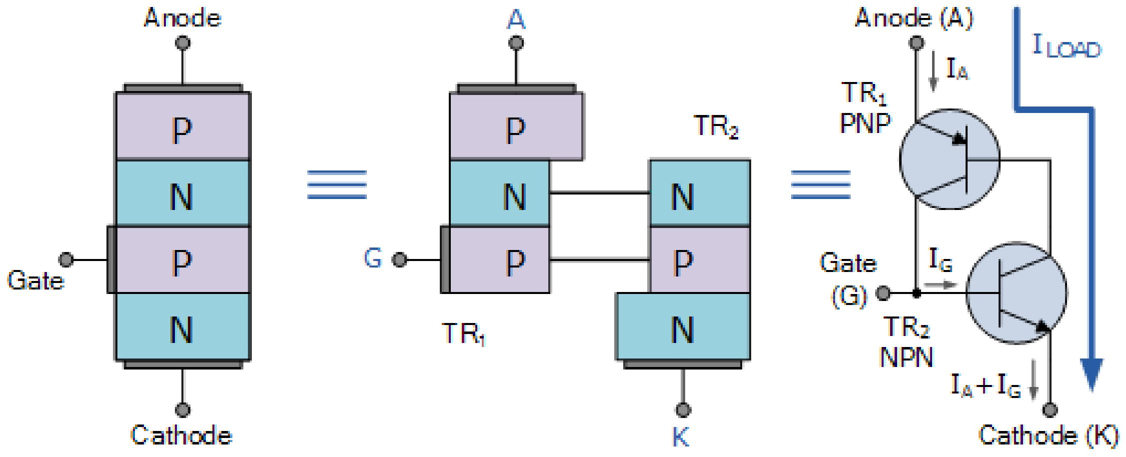
Thyristors - Types, Silicon Controlled Rectifier (SCR), Triode for Alternating Current (TRIAC), Diode for Alternating Current (DIAC), Operating Principle, Parameters/Data, VI Characteristics, Applications

Integrated Circuit (IC) - 555 Timer IC and Its Modes of Operation, Types of ICs (Analog, Digital, and Mixed-Signal), IC Packaging/Pinout/Bases, Applications

Electronic Circuit Design Process - Conceptual Design, Circuit Simulation, Schematic/PCB Design, Development and Prototyping Methods (Breadboard, Veroboard, PCB), CAD Tools, Do-It-Yourself (DIY) PCB Process

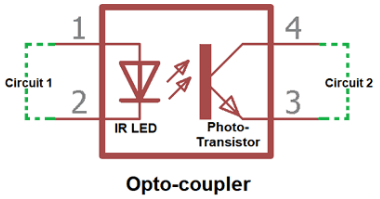

Cathode
Gate
N type (cathode)
P type (gate)
N type
P type (anode)
Anode
(Base Plate)
Mounting
Stud
Anode
(case)

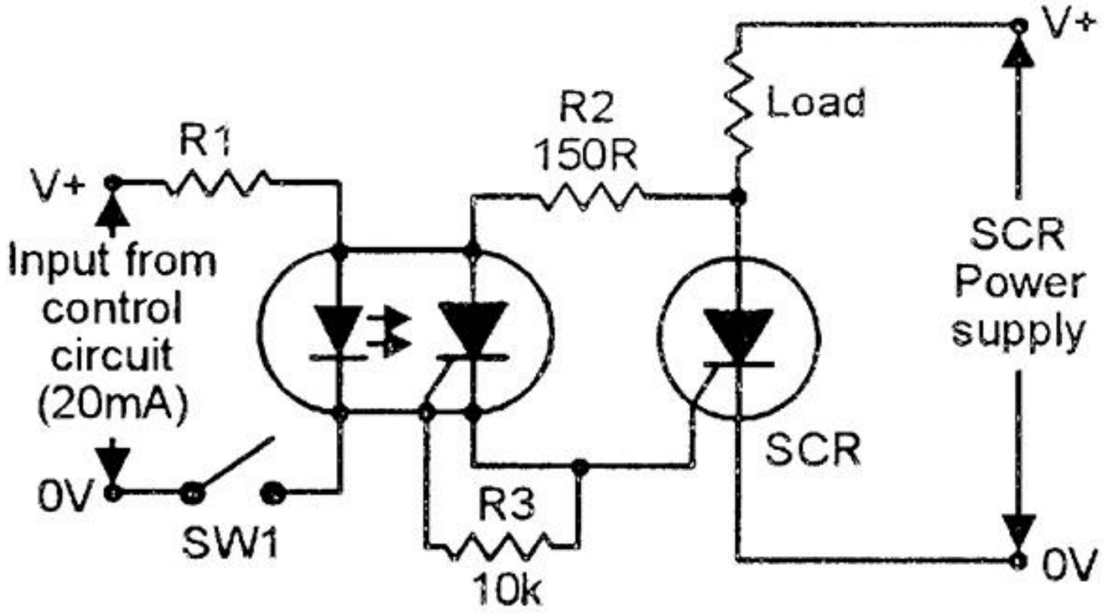

Cathode
Gate
k
g
a
Anode


Anode
P
N
P
N
Gate
Cathode
A
P
N
P
G
TR$_1$
TR$_2$
N
P
N
K
Anode (A)
I_A
I_{LOAD}
TR$_1$
PNP
Gate
(G)
I_G
TR$_2$
NPN
$I_A + I_G$
Cathode (K)

What are Thyristors?

Thyristors are a family of **semiconductor devices** used to **control and switch high voltage and current**, primarily in **power electronics**. They act as **electronic switches** and are typically used in **AC/DC power control**, **motor drives**, **lighting dimmers**, and **inverters**.





Basic Structure & Working Principle

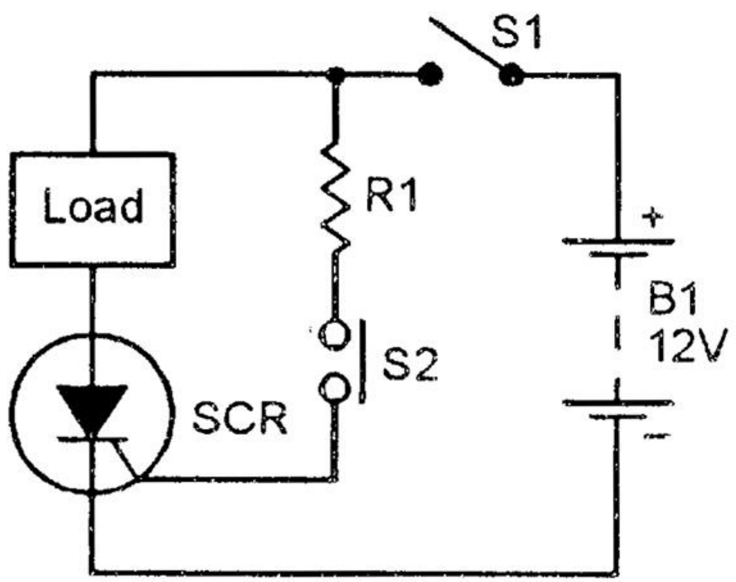
Four layers of alternating P-type and N-type materials (PNPN)

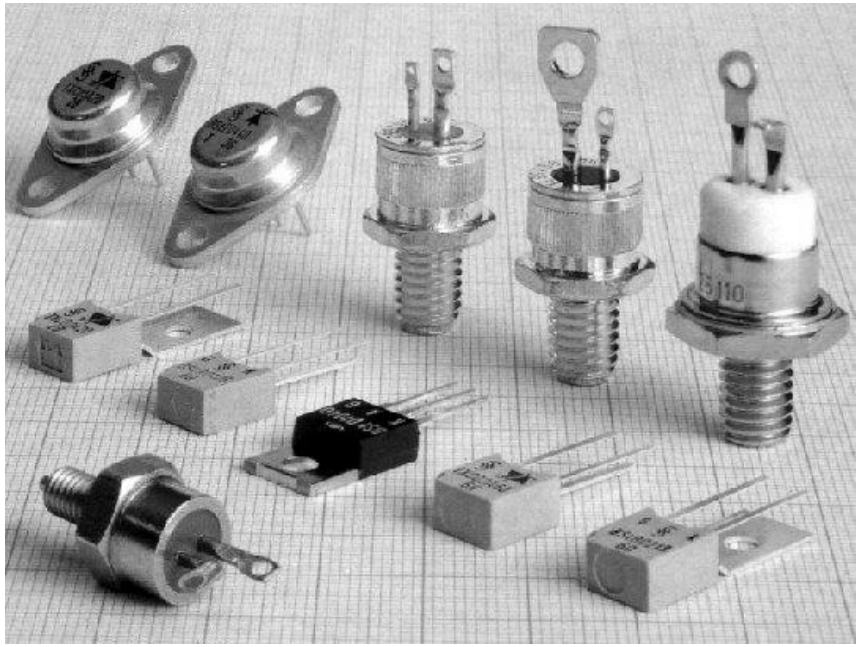
Three terminals: **Anode (A), Cathode (K), Gate (G)** (except for Diac)

It behaves like a **latching switch**:

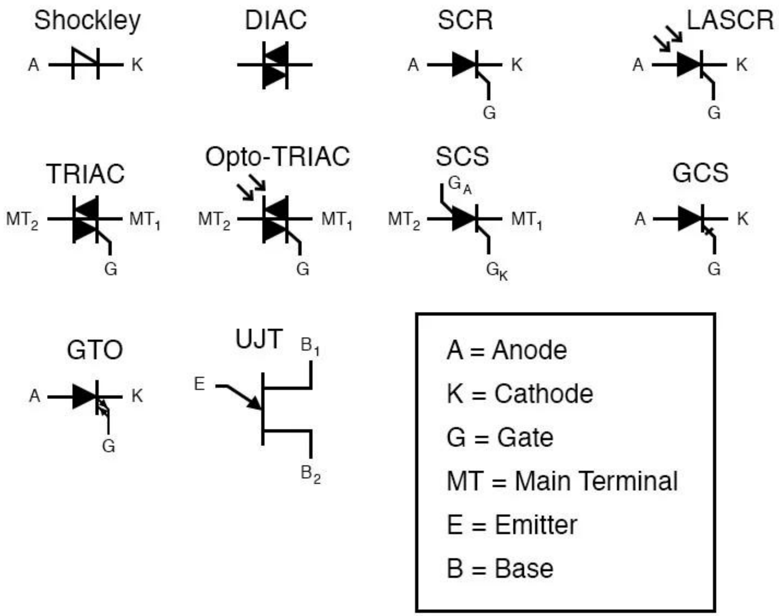
- **OFF** state: Blocks current until triggered.
- **ON** state: Conducts current after triggering and stays ON even if the gate signal is removed (as long as current flows).

Triggered by a **small gate current and** continues to conduct as long as the **main current is above the holding current**.



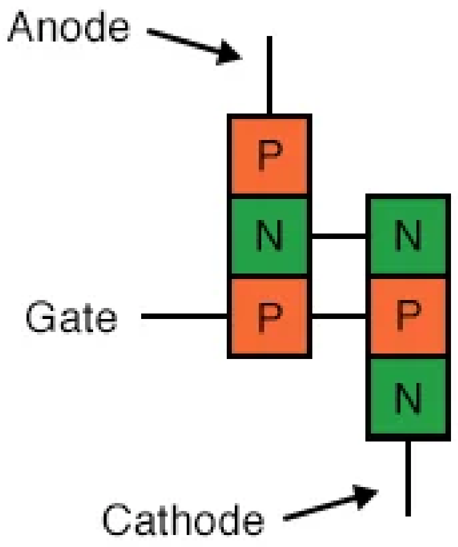


Device	Main Use	Key Characteristics
SCR (Silicon Controlled Rectifier)	DC/AC control	Unidirectional; triggered by gate
Triac	AC dimmers, fan speed control	Bidirectional; gate-controlled
Diac	Triggering Triacs	Bidirectional; no gate
GTO (Gate Turn-Off Thyristor)	High-power converters	Can be turned off via gate
IGCT (Integrated Gate-Commutated Thyristor)	Industrial drives	Faster switching, used in modern high-power systems

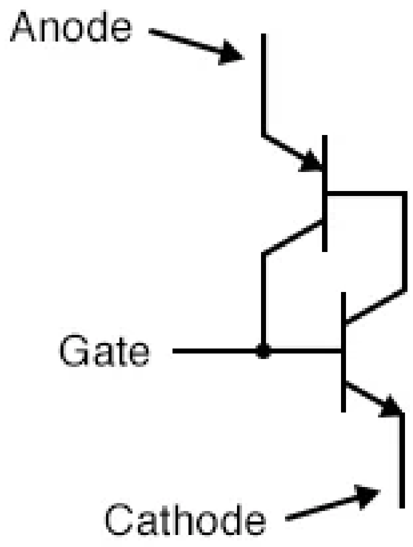


Silicon Controlled Rectifier (SCR)

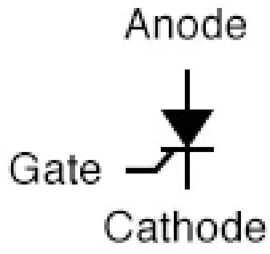
An **SCR** is a **four-layer**, **three-terminal**, **unidirectional semiconductor device** used to control **high voltage and current** in power electronics. It is the most basic and widely used member of the **thyristor family**.



Physical diagram



Equivalent schematic



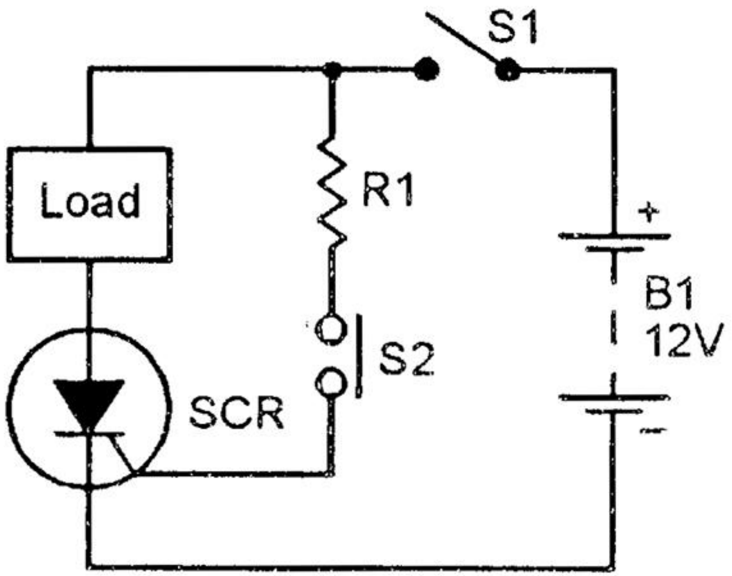
Schematic symbol

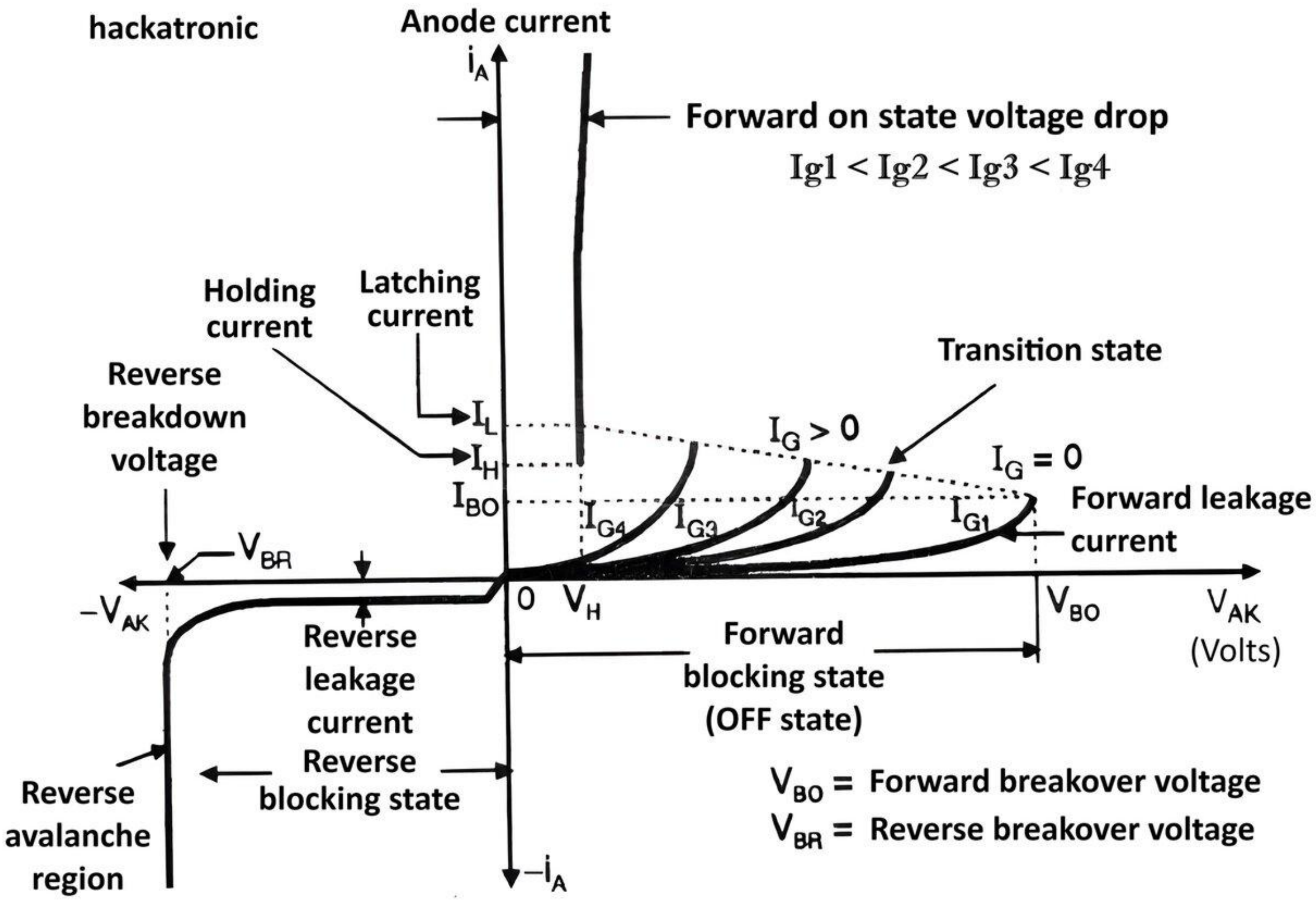
SCR Working Principle

OFF State: When a voltage is applied from **anode to cathode**, the SCR **blocks current**—like an open switch. It stays OFF until a **trigger pulse** is applied to the **gate**.

ON State (Triggered): A **small current at the gate** turns the SCR ON. Once turned ON, it **latches** and continues to conduct **even if the gate current is removed**, as long as the anode current stays above a certain threshold (called **holding current**).

Turn OFF: In DC circuits: You must **reduce the anode current to zero or interrupt the power supply**. In AC circuits: SCR **automatically turns OFF** at every zero-crossing of the AC waveform (natural commutation).





V-I Characteristics of SCR

- **Forward Blocking Region:** No conduction without gate signal.
- **Forward Conduction Region:** Starts conducting after gate pulse.
- **Reverse Blocking Region:** Blocks current like a diode in reverse.

Advantages of SCRs

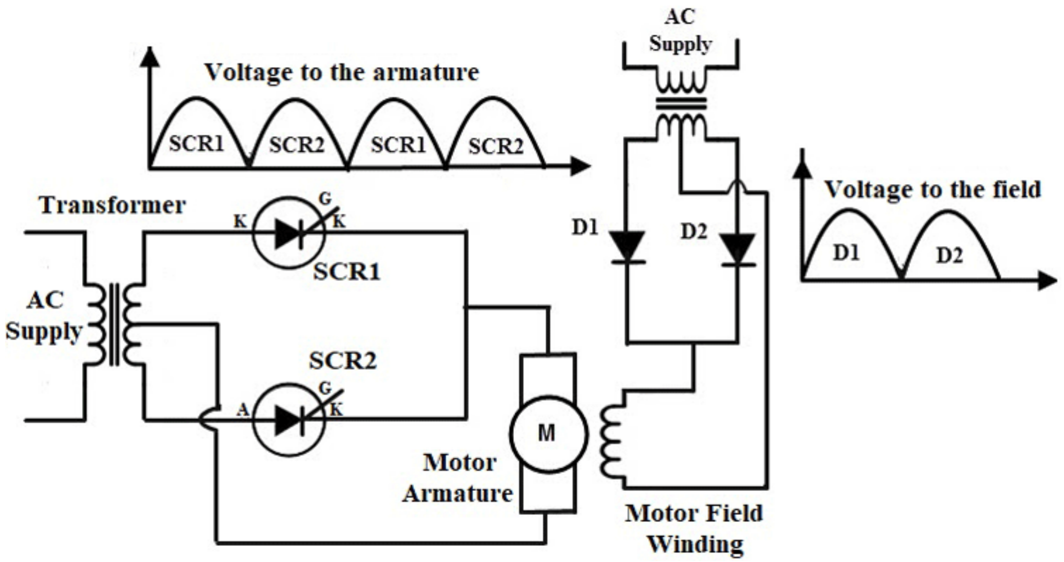
- Can handle **very high voltage and current**
- **Gate requires very little power** to control large loads
- **Reliable** and **compact**
- Provides **isolation** between control and power sides

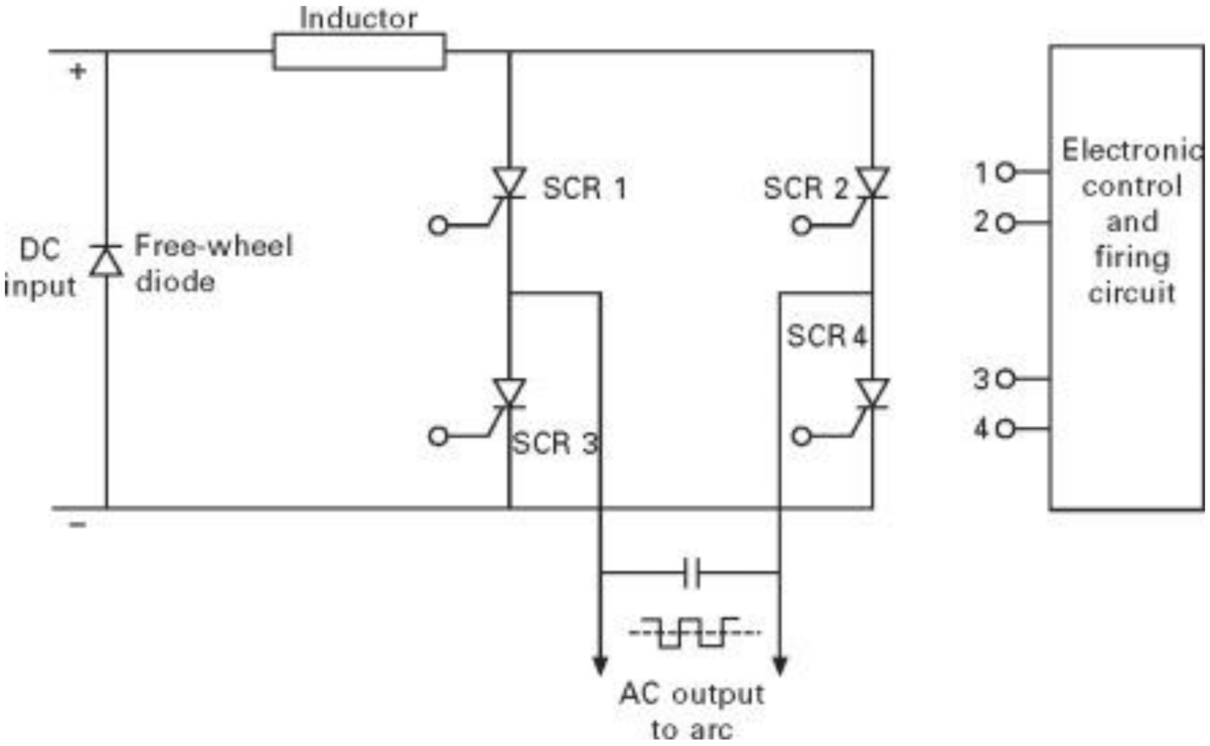
Limitations of SCRs

- **Unidirectional** (only conducts in one direction)
- Cannot be turned off using the gate
- Not suitable for **high-frequency** switching
- Needs external circuitry to turn OFF in DC circuits

Applications of SCRs

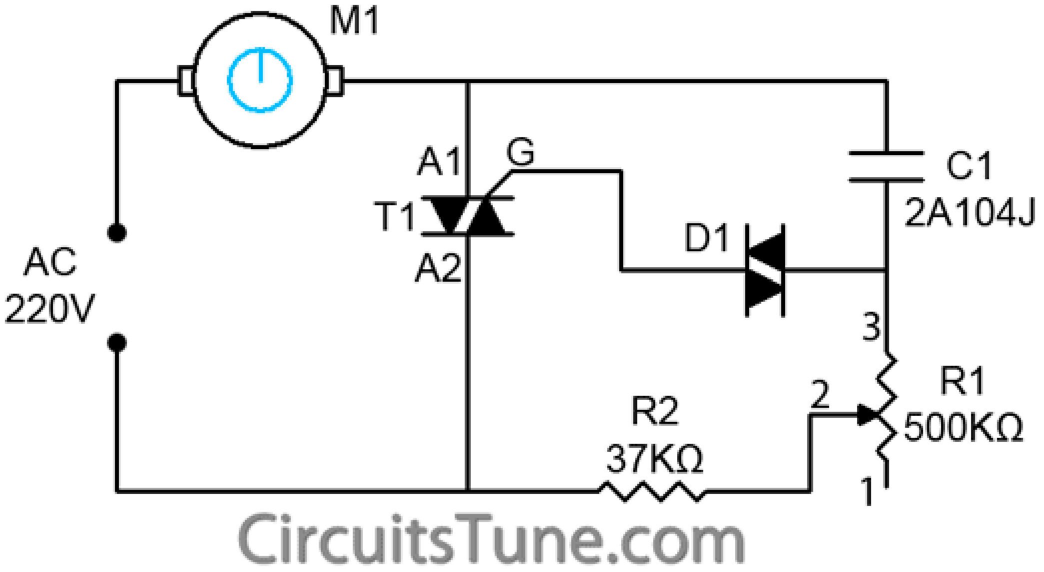
Application	Role of SCR
Light dimmers	Phase control in AC
Motor control	Speed regulation
Battery chargers	Controlled rectification
Crowbar circuits	Overvoltage protection
Inverters & UPS	Power conversion
Controlled rectifiers	AC to DC conversion with control



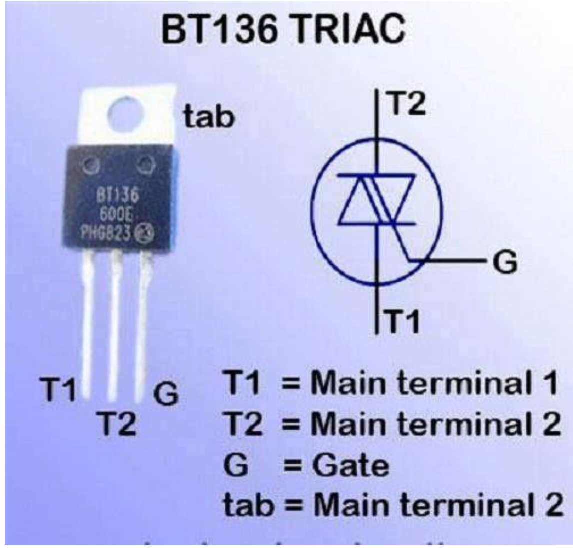


Triode for Alternating Current (TRIAC)

A **Triac** (Triode for Alternating Current) is a **bidirectional** semiconductor **switching device** that can control **AC power**. It belongs to the **thyristor family** and is widely used in **light dimmers, fan regulators, motor speed controllers**, and other AC switching applications.

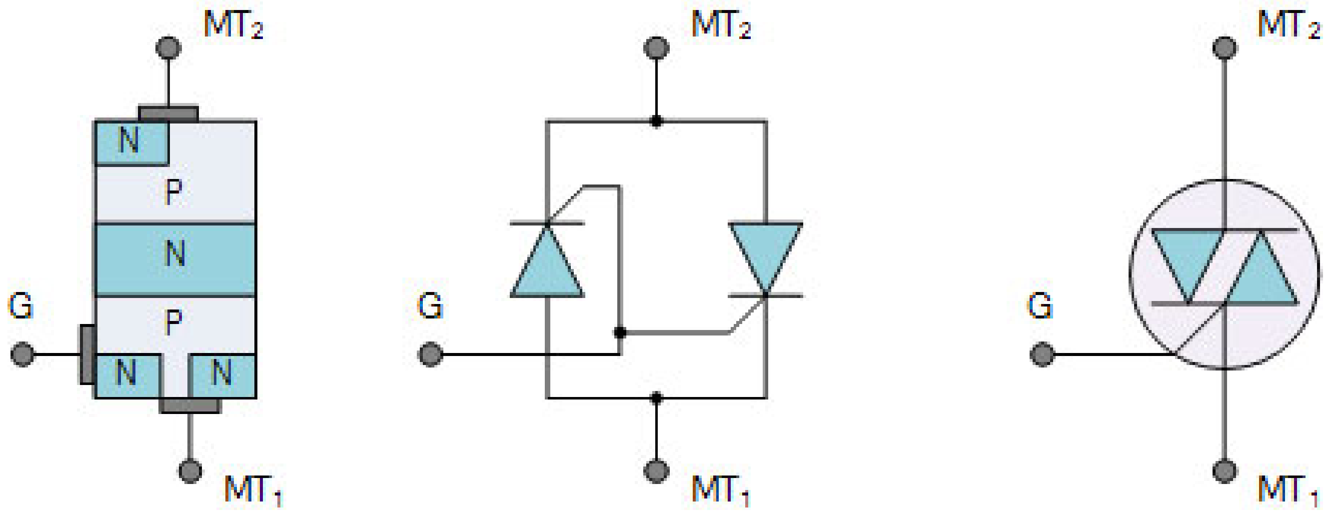


Structure and Terminals of TRIAC



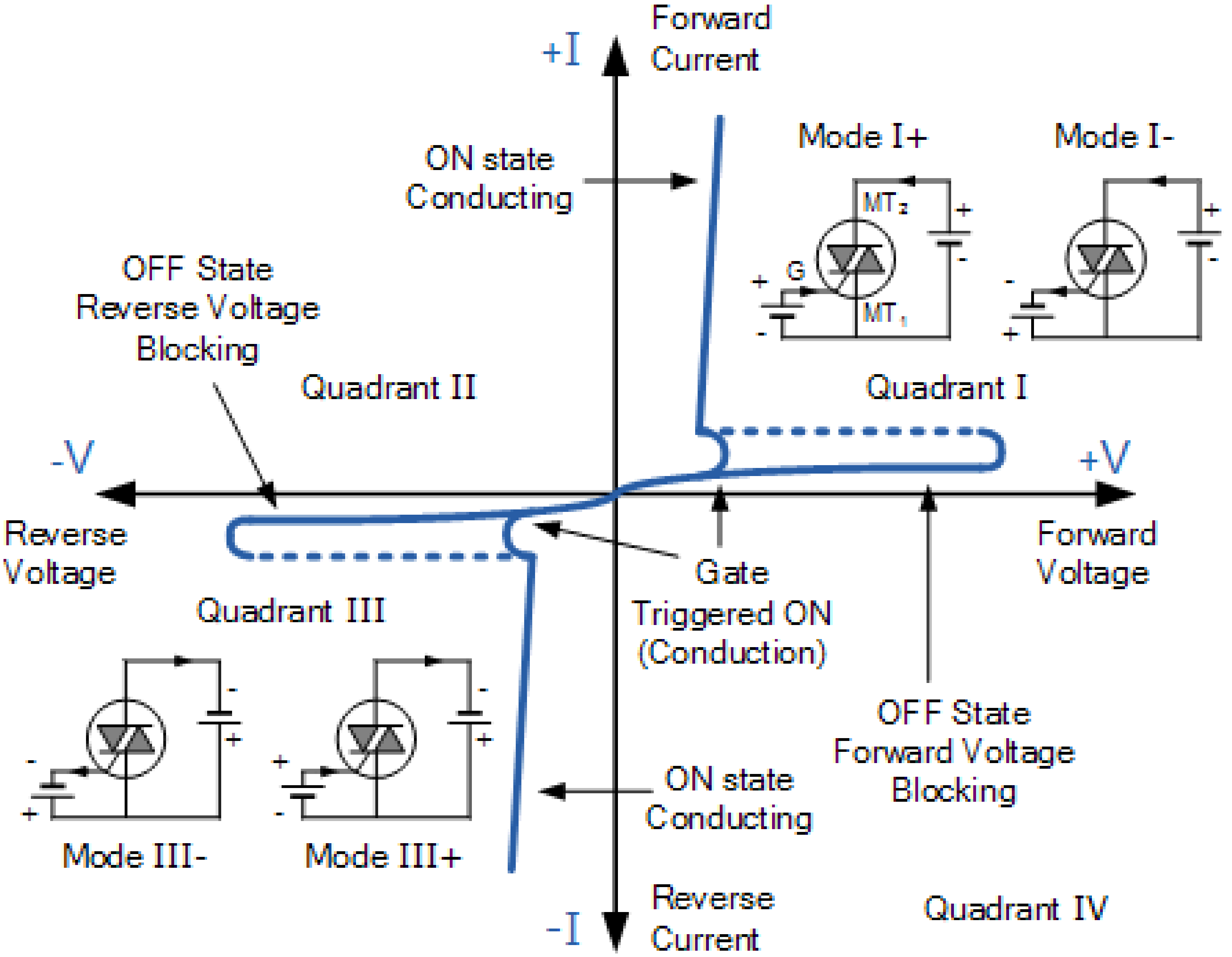
A **Triac** is equivalent to **two SCRs connected in anti-parallel** with a shared gate. It can conduct **in both directions** when triggered.

- **Main Terminals:**
  - **MT1 (Main Terminal 1)**
  - **MT2 (Main Terminal 2)**
- **Gate (G):** Used to trigger the device ON



Feature / Parameter	BT136	BT137	BT138	BT139
Max RMS Voltage	600V	600V	600V / 800V options	600V / 800V options
Max RMS Current (IT(RMS))	**4A**	**8A**	**12A**	**16A**
Gate Trigger Current (I_GT)	~5 to 50 mA (typical)	~10 to 50 mA	~10 to 50 mA	~10 to 50 mA
Max Surge Current (ITSM)	~25A	~65A	~100A	~150A
Package	TO-220AB	TO-220AB	TO-220AB	TO-220AB
Typical Application	Low-power AC loads	Medium-power loads	High-power AC loads	Industrial AC loads
Cost	Low	Moderate	Higher than BT137	Highest in this group

TRIAC VI Curve



Advantages of TRIAC

- **Bidirectional control** with a single device
- **Reduces component count** vs. two SCRs
- Works efficiently in **AC power control**
- Triggered by **positive or negative gate current**

Disadvantages of TRIAC

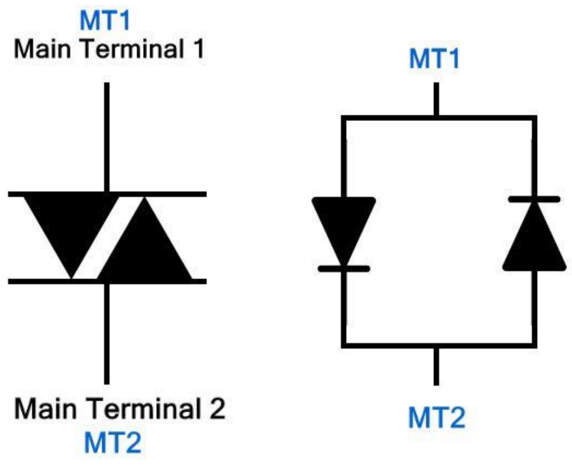
- **Sensitive to noise**, especially at low gate currents
- Less rugged than SCRs; can **fail with high inrush currents**
- Not suitable for **high-frequency** applications

Applications of TRIAC

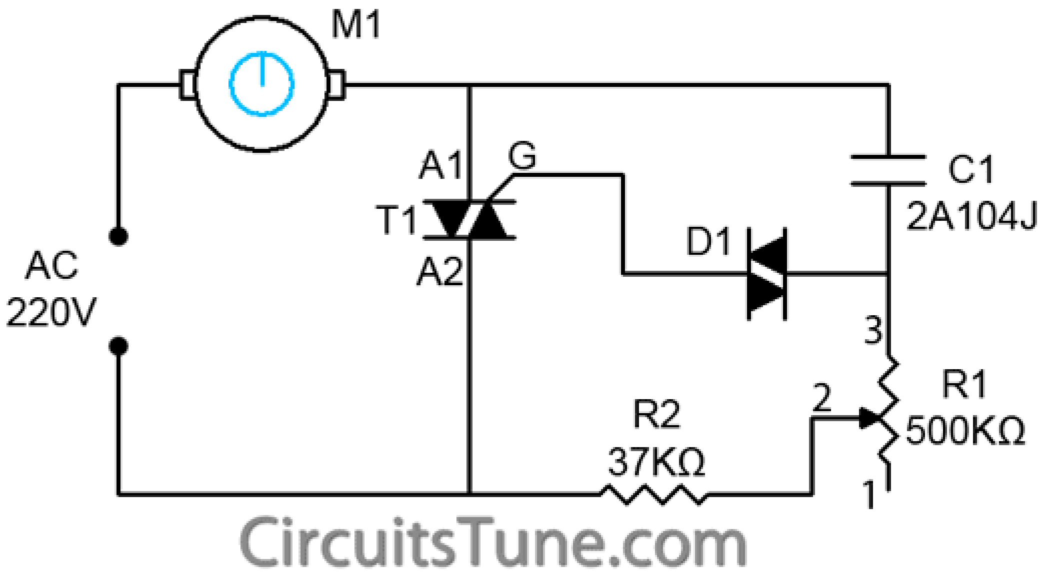
Application	Function
AC light dimmers	Vary brightness by phase control
Fan regulators	Adjust fan speed
Heater controls	Regulate power to heating elements
Solid-state relays	Switch AC loads electronically
Motor speed control	Adjust induction motor speed (low power)

Diode for Alternating Current (DIAC)

A **DIAC** (Diode for Alternating Current) is a **bidirectional trigger device** that conducts current **only after a certain voltage threshold** (called **breakover voltage**) is reached in either direction. It is mainly used to **trigger TRIACs** in AC power control circuits like **light dimmers**, **fan regulators**, and **heater controllers**.



DIAC Symbol



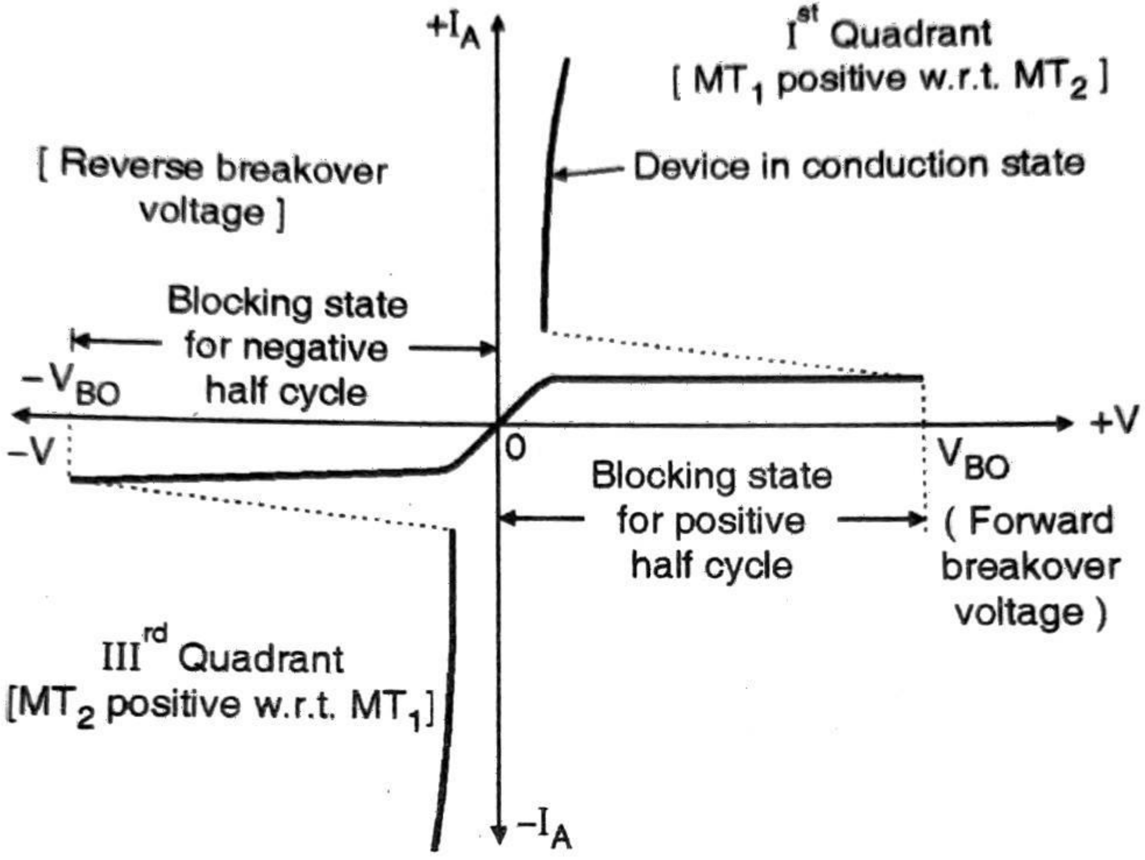
Working Principle of DIAC

In the **OFF state**, the DIAC blocks current like an open circuit.

When the **voltage across the DIAC exceeds the breakover voltage (typically 30–40V)**, it suddenly switches ON and allows **current to flow**.

Once conducting, the DIAC continues until the current drops below the **holding current**, then switches OFF.

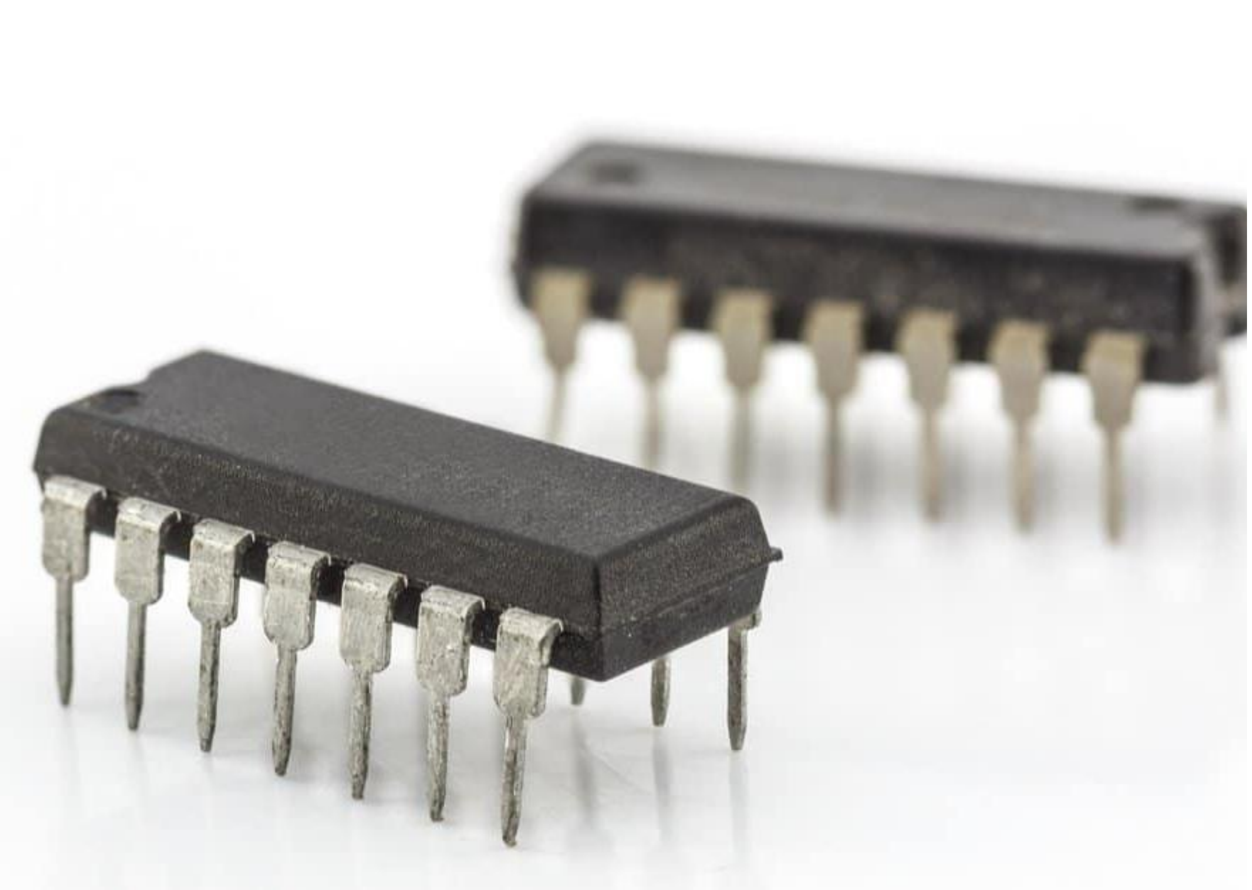
This **sharp triggering** makes it ideal for firing **TRIACs** reliably.

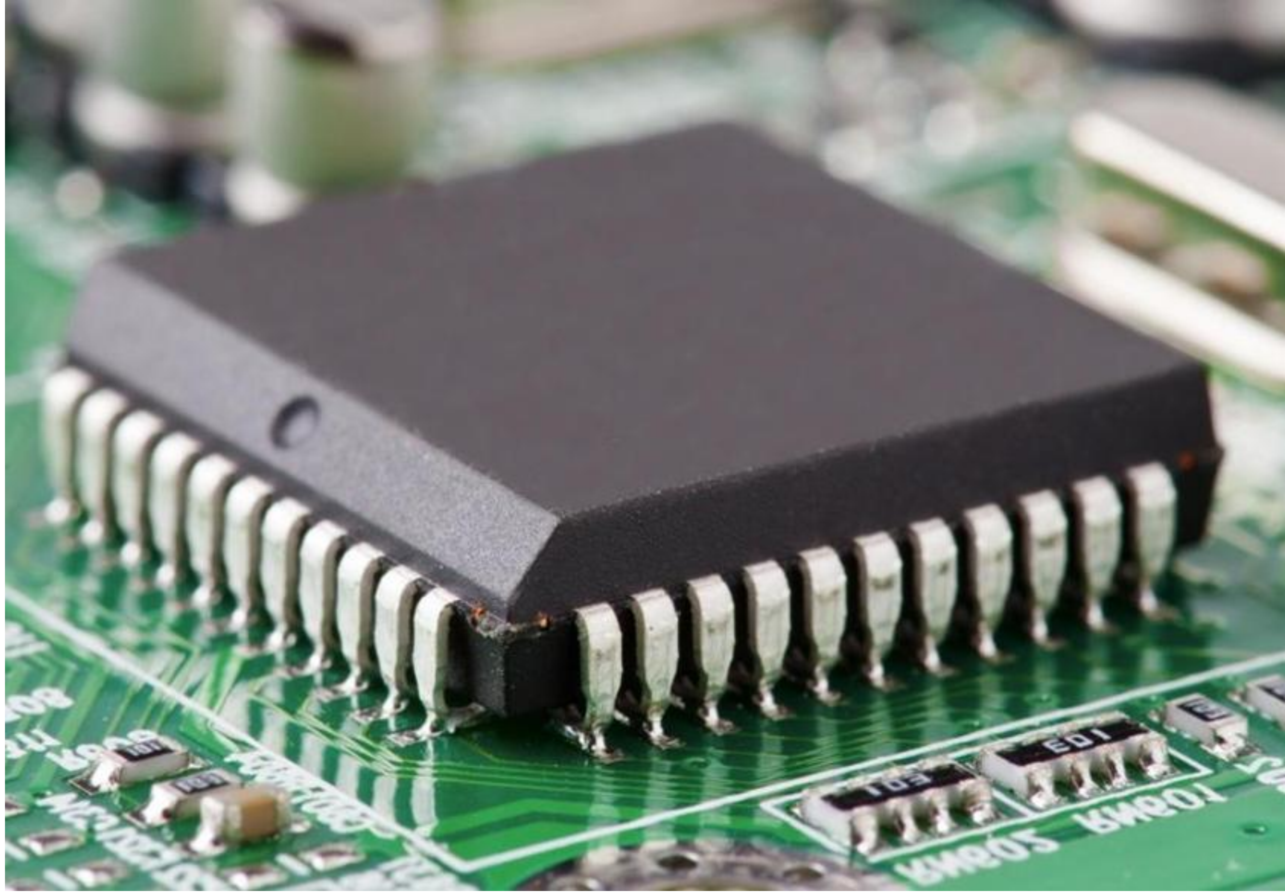


VI characteristics of DIAC

Comparison Table: BJT vs FET vs Thyristor

Feature / Aspect	BJT (Bipolar Junction Transistor)	FET (Field Effect Transistor)	Thyristor (e.g., SCR, TRIAC)
Control Type	**Current-controlled** device	**Voltage-controlled** device	**Latch-controlled** (via gate pulse)
Terminals	Emitter, Base, Collector	Source, Gate, Drain	Anode, Cathode, Gate
Switching Type	Linear and switching	Linear and switching	Mainly switching (ON/OFF only)
Conduction	Continuous (controlled by base current)	Continuous (controlled by gate voltage)	**Latching**: once ON, stays ON until current drops
Triggering Signal	Base current (~mA)	Gate voltage (~V)	Gate pulse (~mA, momentary)
Directionality	Unidirectional	Unidirectional (MOSFET) or bidirectional (JFET)	SCR: UnidirectionalTRIAC: Bidirectional
Turn-OFF Control	Controlled via base current removal	Controlled via gate voltage removal	Needs **external commutation** or AC zero-crossing
Speed (Switching Time)	Fast (µs range)	Very fast (ns–µs range)	Slower (µs–ms range)
Power Handling	Low to moderate	Low (MOSFET) to moderate (IGBT)	**High power** (SCRs, TRIACs)
Applications	Amplifiers, logic switches, drivers	Logic ICs, switching regulators, power conversion	AC power control, dimmers, motor drives
Gain	Has current gain (β)	No current gain (voltage-controlled)	No gain (switching device)
Input Impedance	Low (base current needed)	High (gate draws little current)	Moderate (depends on gate sensitivity)



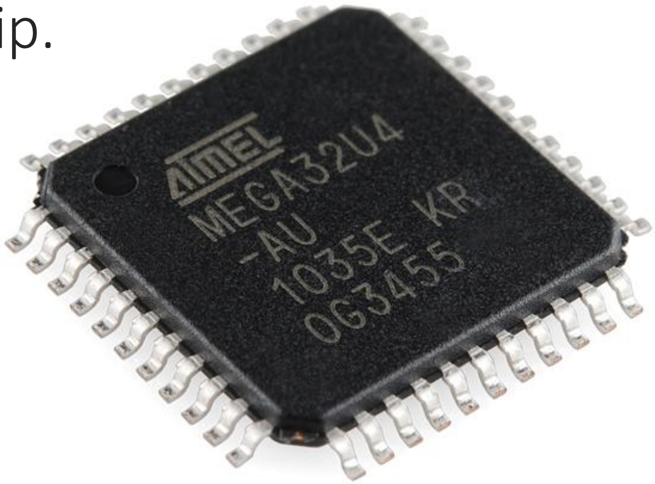


Integrated Circuit (IC)

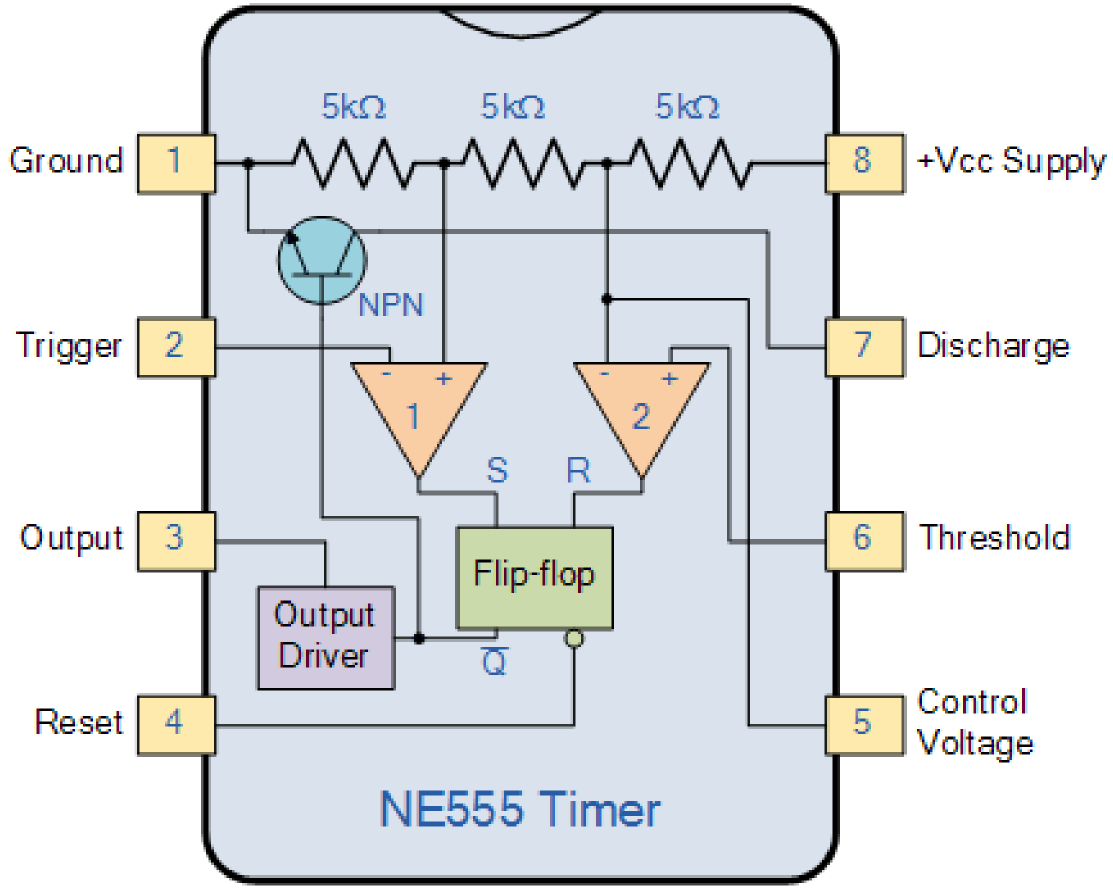
An **Integrated Circuit (IC)** is a small electronic device made of a semiconductor material, usually **silicon**, that contains a large number of **tiny electronic components** such as transistors, diodes, resistors, and capacitors, all fabricated onto a single chip.

Key Features:

- **Miniaturization**: Combines many components into a tiny chip.
- **High reliability**: Fewer soldered joints than discrete circuits.
- **Low cost (in mass production)**.
- **Low power consumption**.
- **Fast performance**.



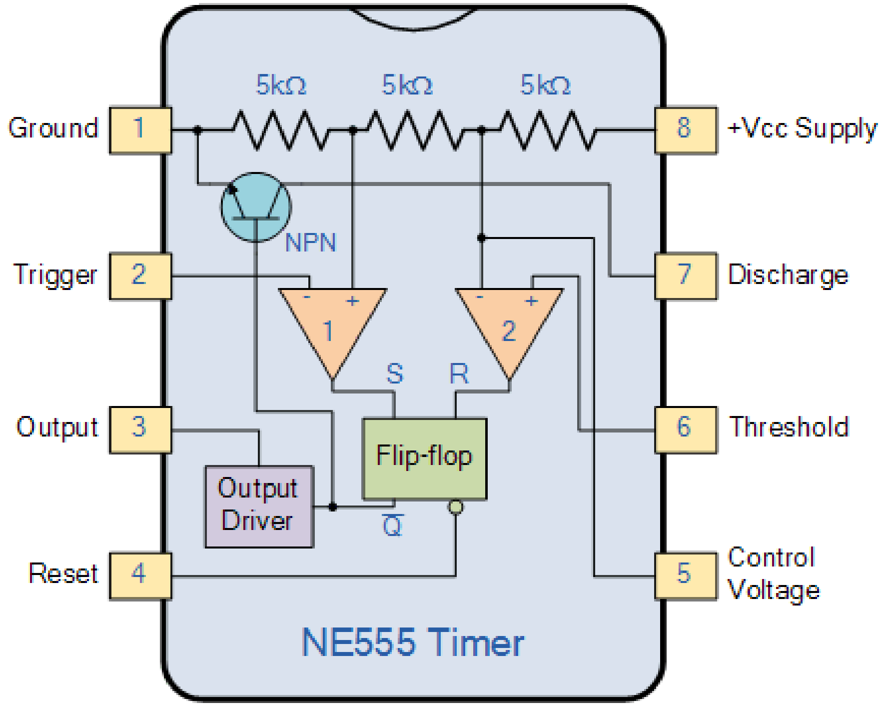
Example: 555 Timer IC



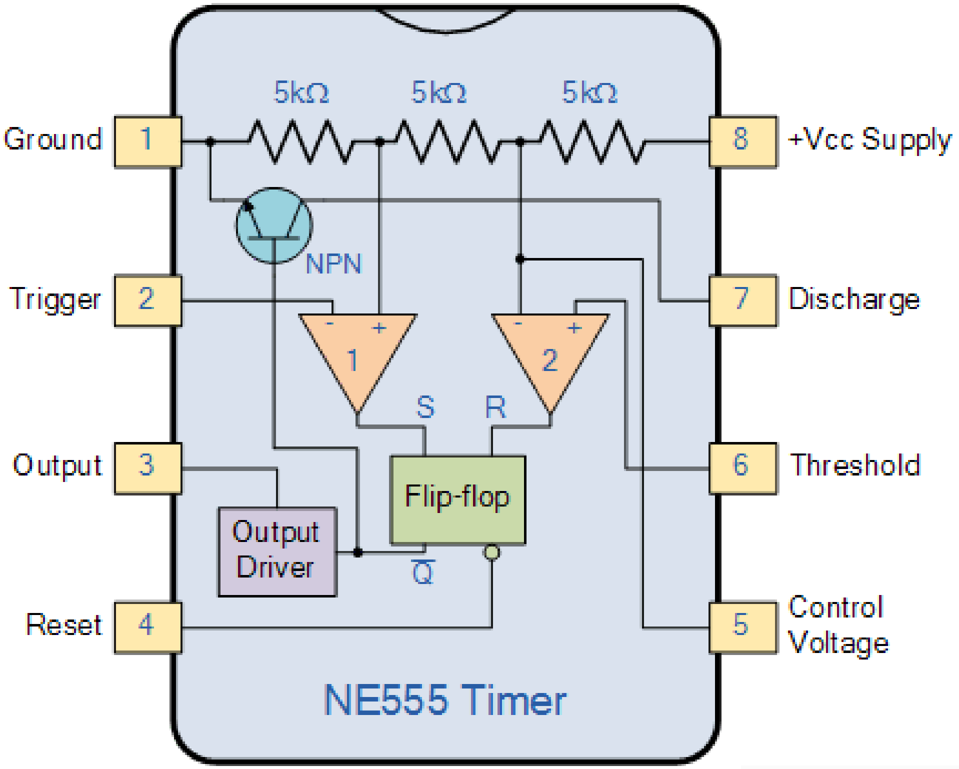
555 Timer IC

The **555 Timer IC** is one of the most popular and versatile integrated circuits ever made. It was introduced by **Signetics in 1972** (now part of ON Semiconductor) and is widely used in **timing, pulse generation, and oscillator applications**.

Pin	Name	Description
1	GND	Ground (0V)
2	Trigger	Starts the timing cycle (active low)
3	Output	Output signal (HIGH or LOW)
4	Reset	Resets the timer (active low)
5	Control Voltage	Optional control of threshold (usually unused)
6	Threshold	Compares voltage to end timing cycle
7	Discharge	Discharges timing capacitor
8	Vcc	Supply voltage (4.5V to 15V)



555 Timer IC Cont.



Internally, the 555 contains:

- **Two comparators**
- **One SR flip-flop**
- **A discharge transistor**
- **Voltage divider (3 resistors of 5kΩ)**

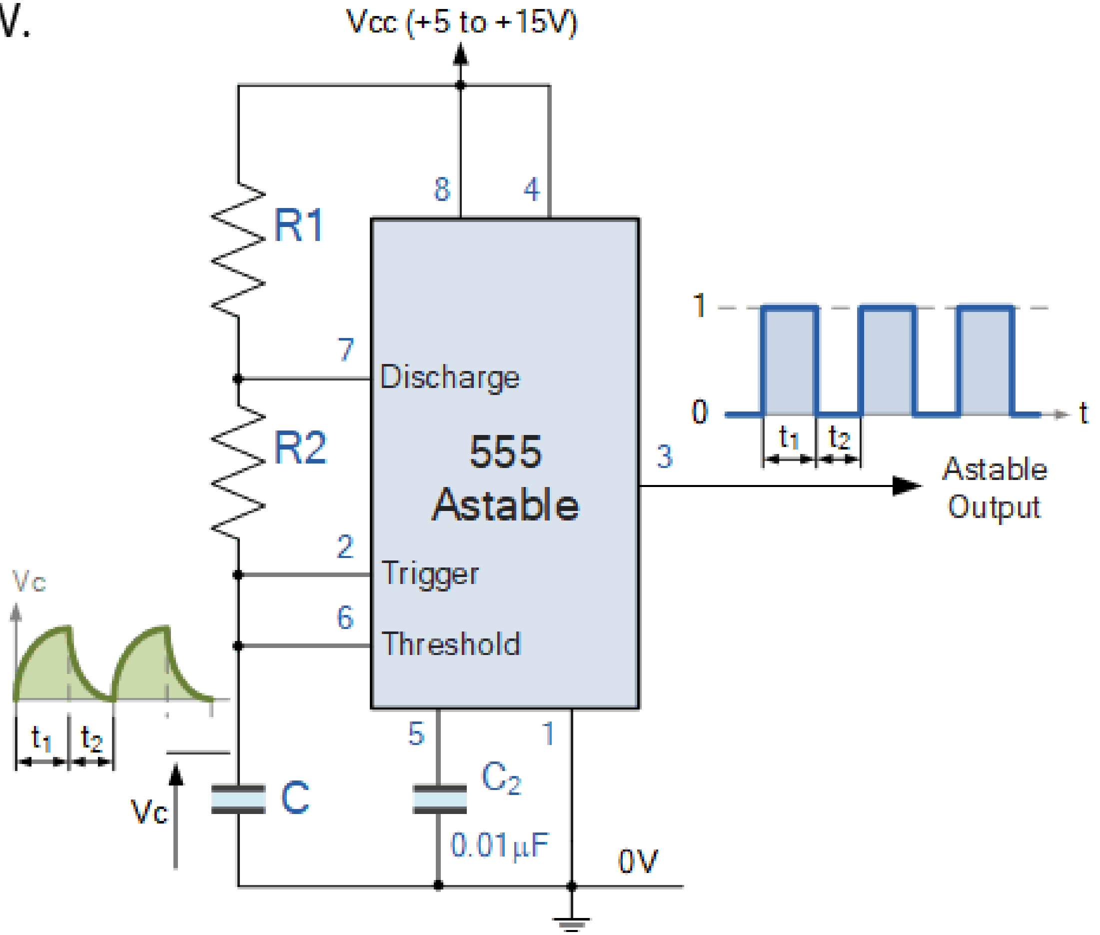
These components allow the IC to **monitor voltages** and **change output states** based on capacitor charging.

555 Timer Modes of Operation: Astable Mode (Oscillator / Pulse Generator)

- No stable state; it continuously toggles HIGH and LOW.
- Used for flashing LEDs, tone generation, clock pulses.

Formula:

- **Frequency (f)** = 1.44 / ((R1 + 2R2) × C)
- **Duty Cycle** ≈ (R1 + R2) / (R1 + 2R2)

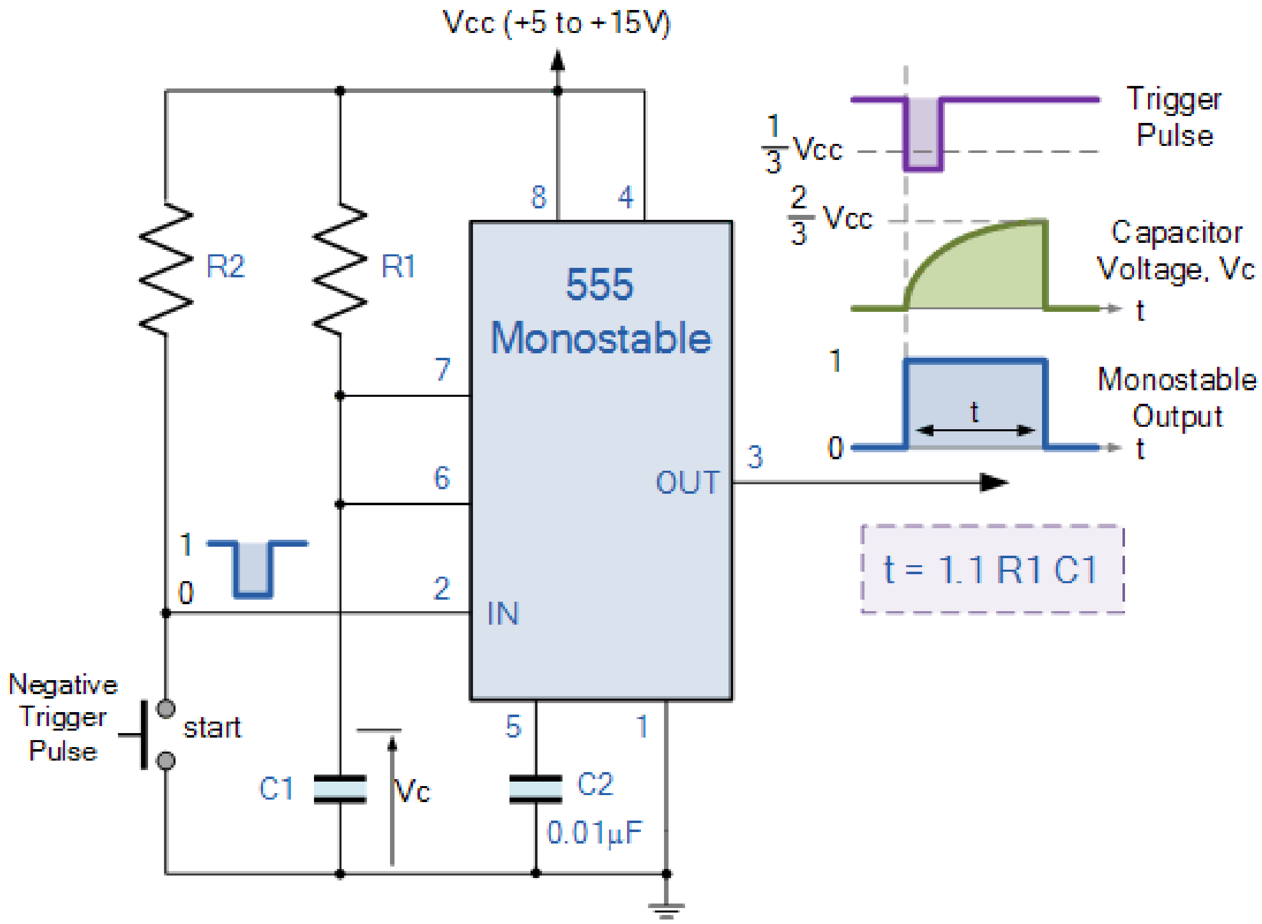


Check https://www.electronics-tutorials.ws/waveforms/555_oscillator.html

555 Timer Modes of Operation: Monostable Mode (One-Shot Pulse)

- Outputs a single pulse when triggered.
- Used in timers, debounce circuits.

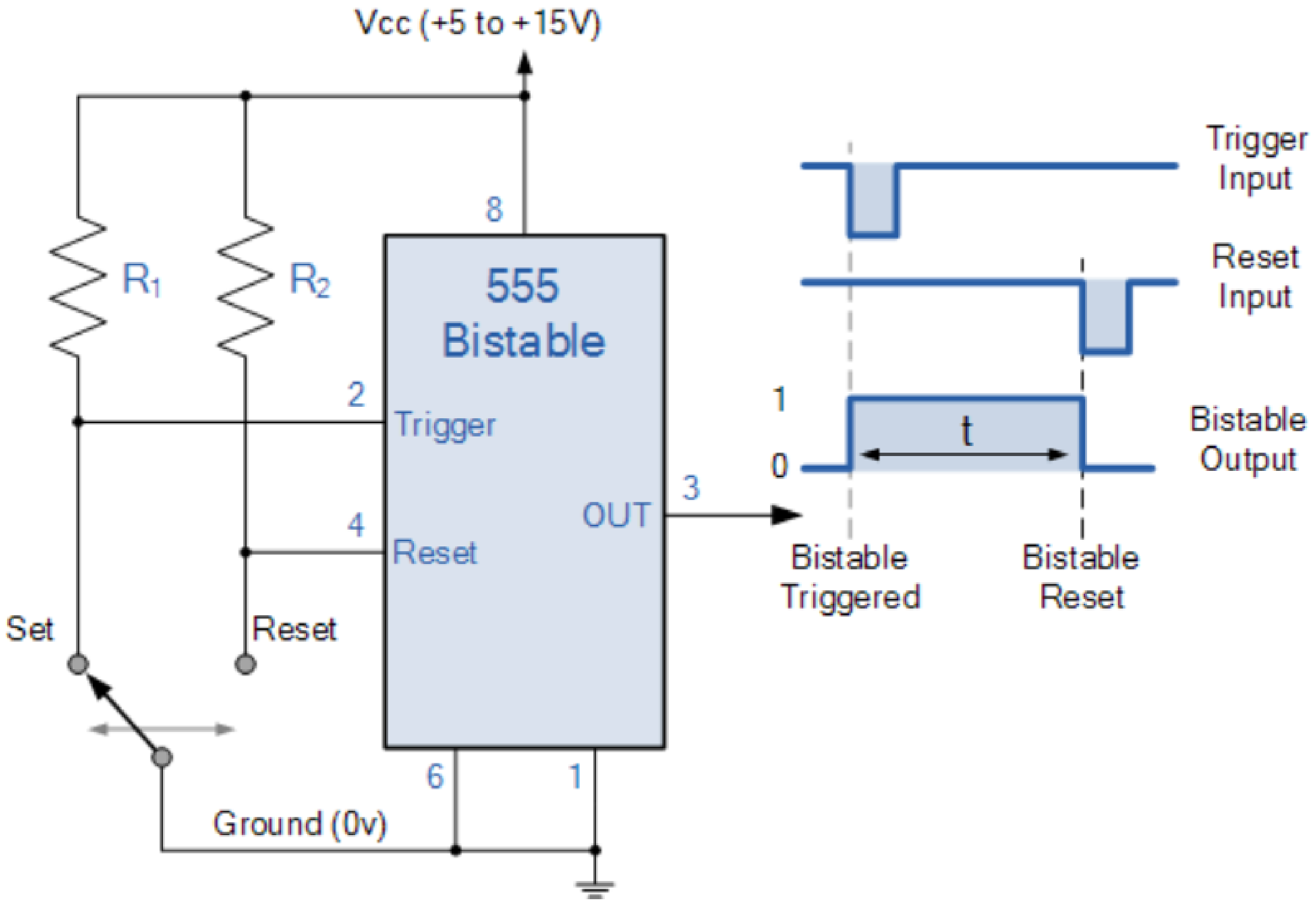
- **Formula:**
- **Pulse Width (T)** = 1.1 × R × C



Check https://www.electronics-tutorials.ws/waveforms/555_timer.html

555 Timer Modes of Operation: Bistable Mode (Flip-Flop)

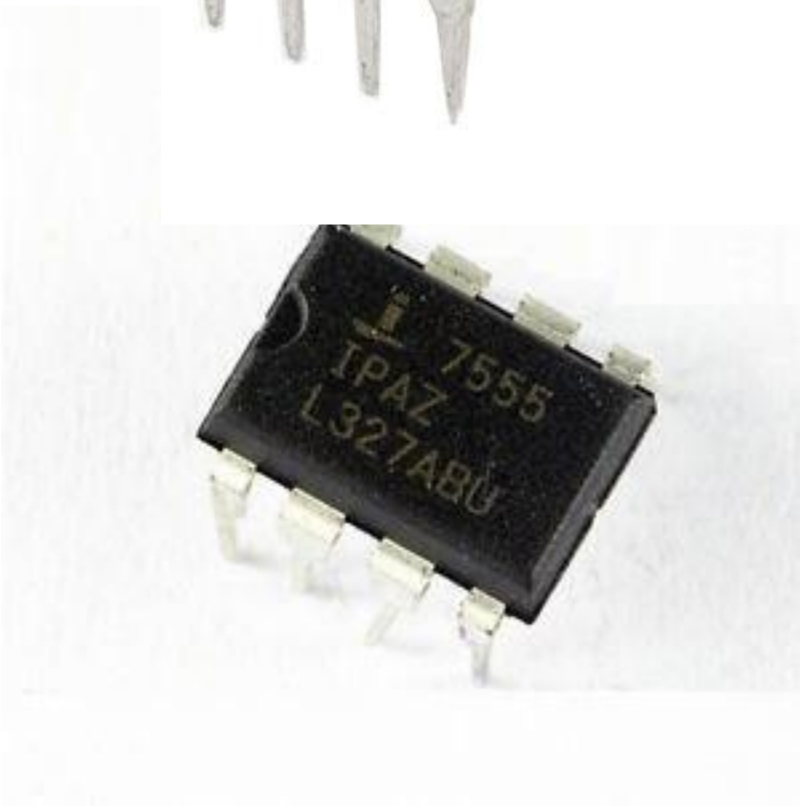
- Toggle between HIGH and LOW based on input triggers.
- Used in switch debouncing and latching circuits



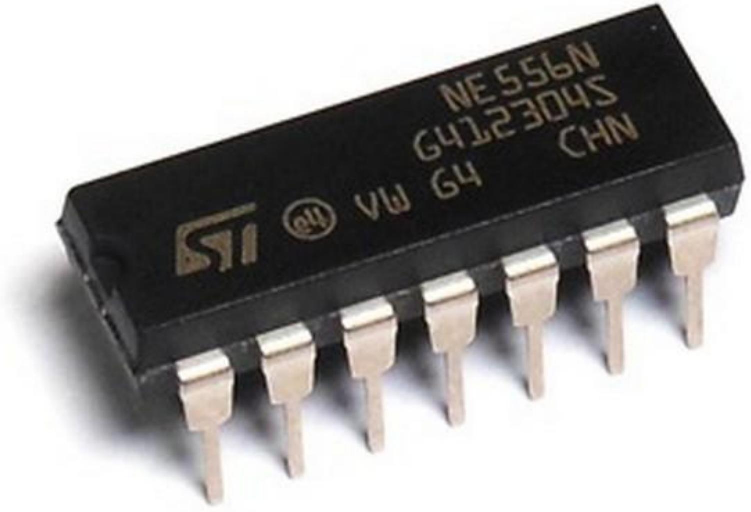
Check https://www.electronics-tutorials.ws/waveforms/555_timer.html

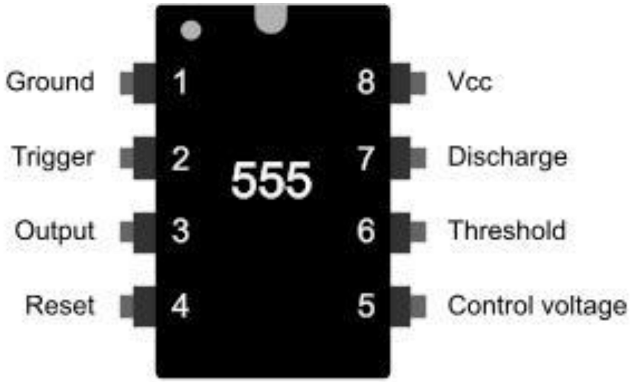
555 Timer Variants

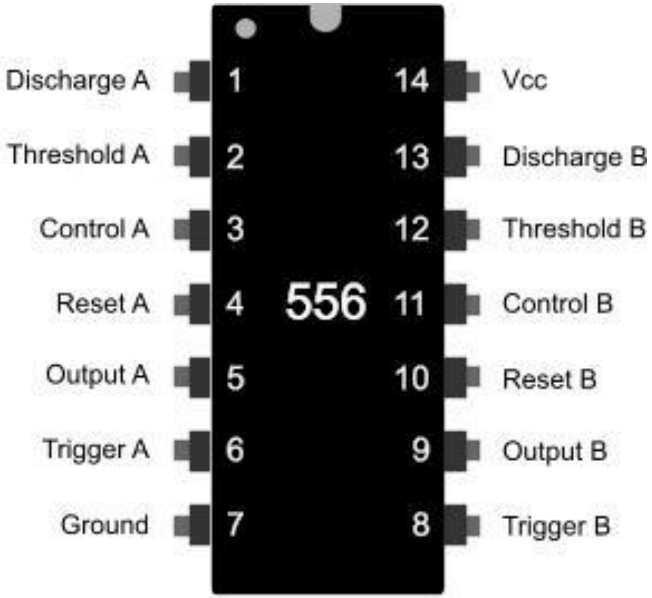
- **NE555** – Standard version
- **LM555** – Texas Instruments version
- **7555** – CMOS version (low power)

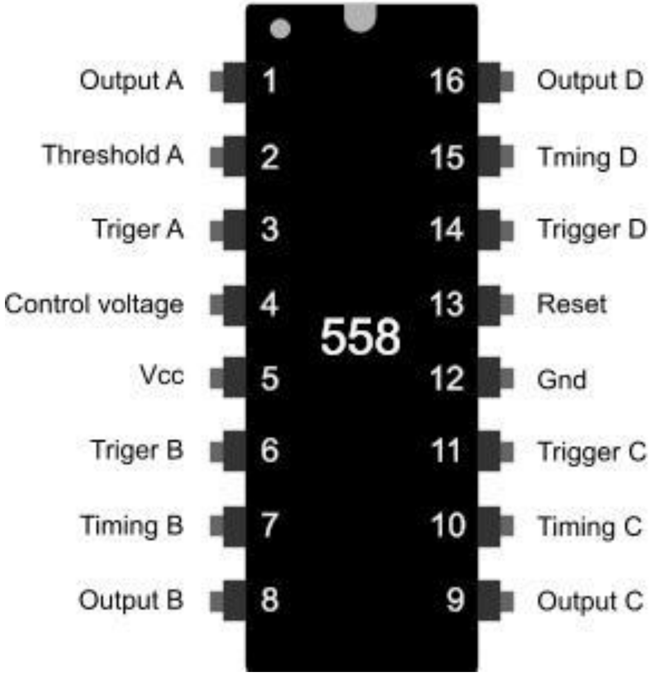


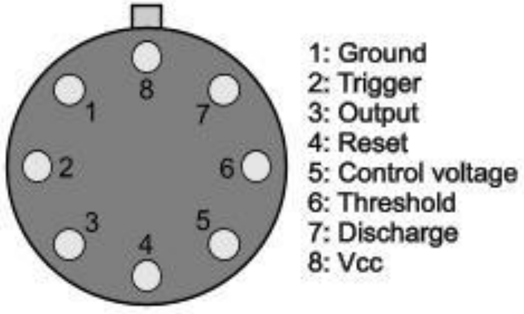
555 Timer Variants Cont.







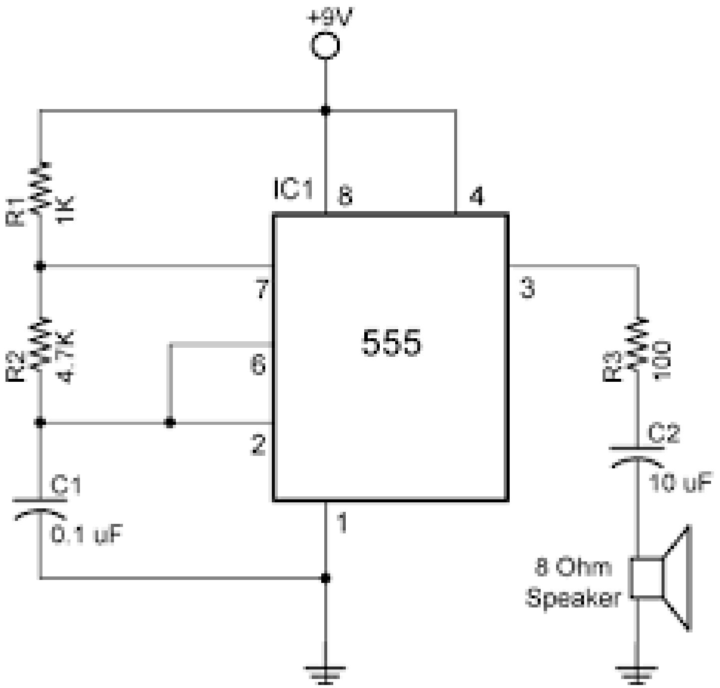


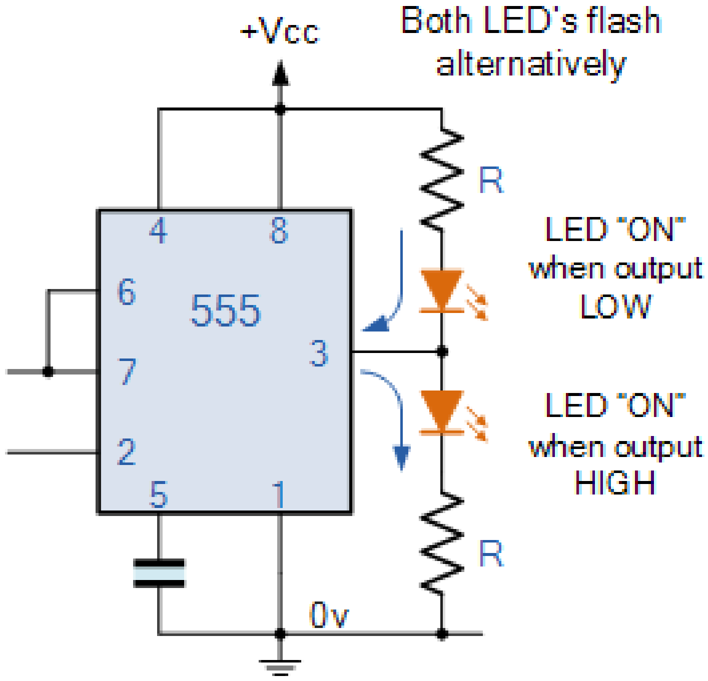


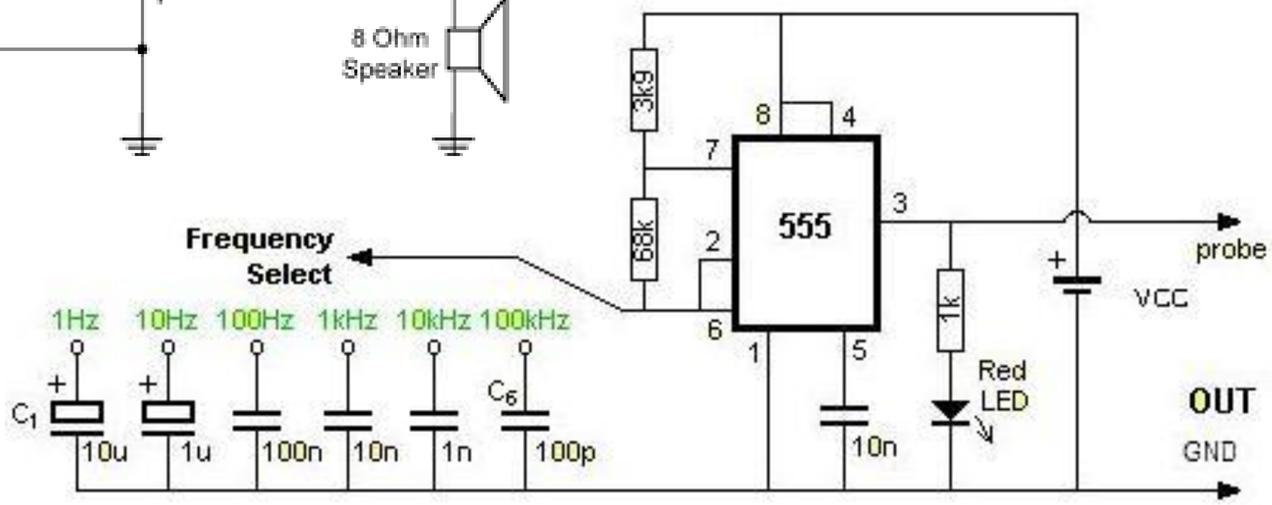
The 'T' Package

Applications of 555 Timer

- LED flashers and blinkers
- Tone generators
- Pulse width modulation (PWM)
- Timers (delay circuits)
- Frequency generators
- Light or temperature sensors
- Power-on delay circuits





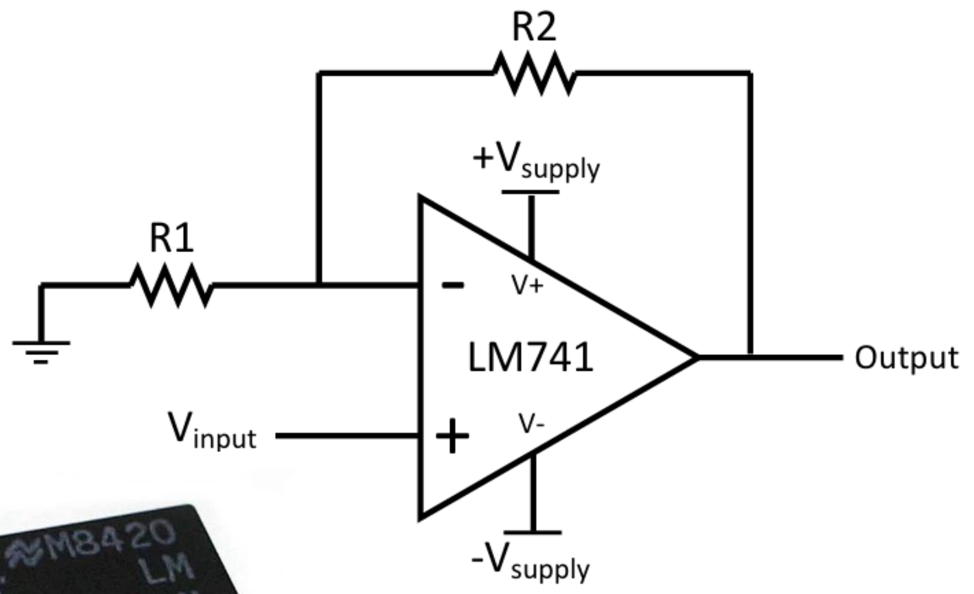


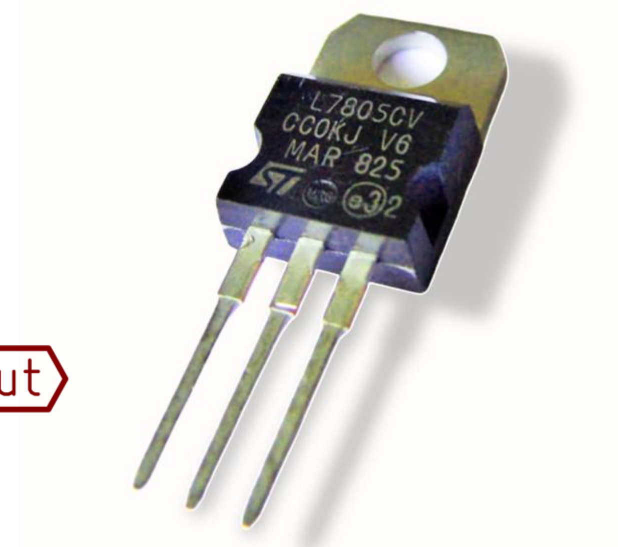
Types of ICs

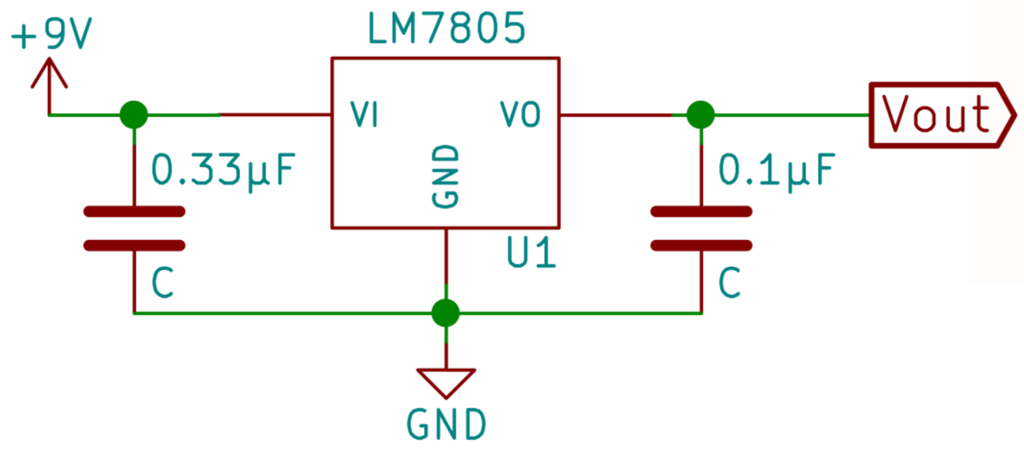
Type	Description	Example
Analog ICs	Handle continuous signals (e.g., audio)	Operational amplifiers (Op-Amps), Voltage regulators
Digital ICs	Handle binary signals (0s and 1s)	Logic gates, Microprocessors, Memory chips
Mixed-Signal ICs	Combine analog and digital functions	Analog-to-Digital Converters (ADC), DACs

Analog ICs

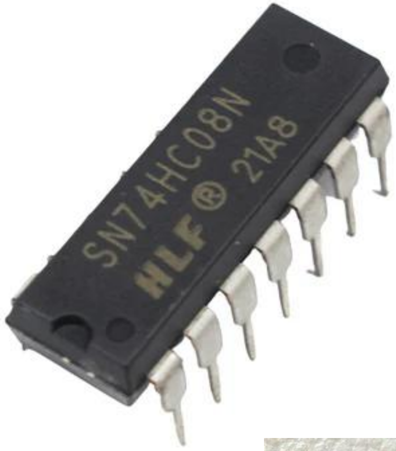
- Work with **continuous signals** (voltages or currents).
- Do signal amplification, filtering, modulation, etc.
- **Examples:**
  - **Op-Amps (e.g., 741)**
  - **Voltage Regulators (e.g., LM7805)**
  - **Comparators**
  - **Audio amplifiers**

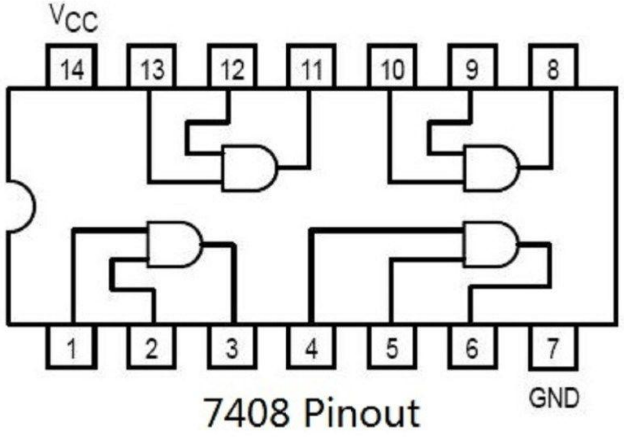




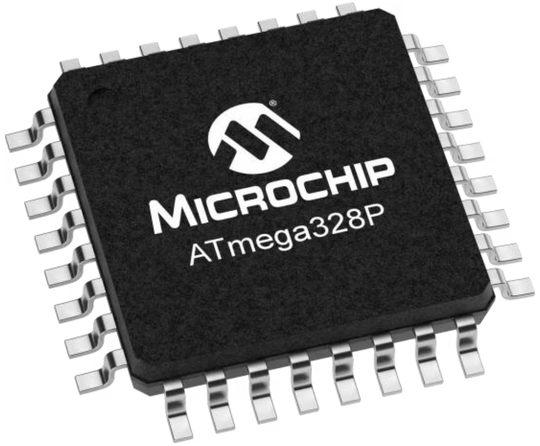


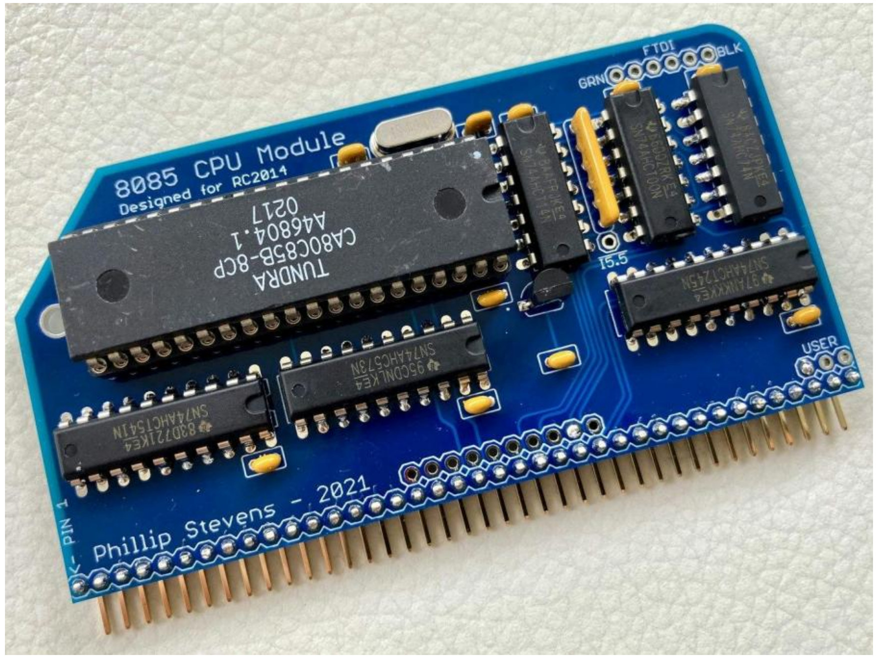
Digital ICs





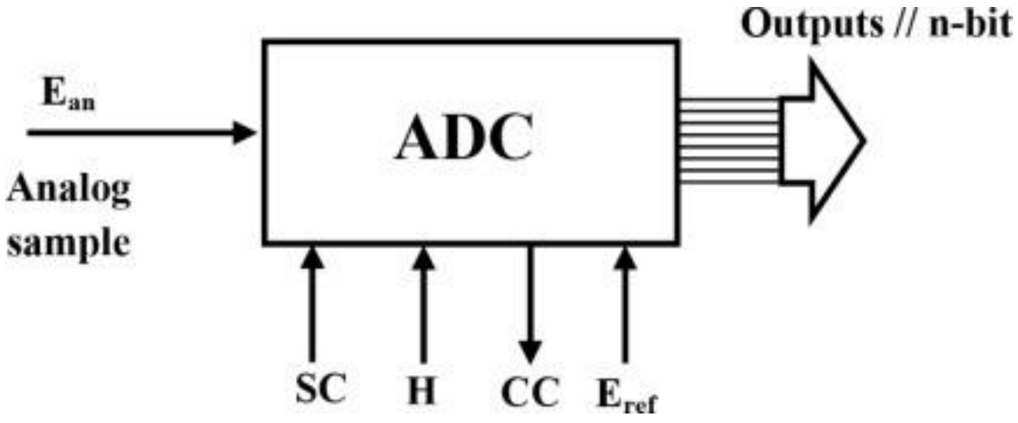
- Work with **discrete signals** (binary: 0 and 1).
- Perform logic operations, data storage, processing.
- **Examples:**
  - **Logic gates (AND, OR, NOT, etc.)**
  - **Flip-flops, Counters, Multiplexers**
  - **Microprocessors (e.g., Intel 8085)**
  - **Microcontrollers (e.g., ATmega328)**
  - **Memory ICs (RAM, ROM)**

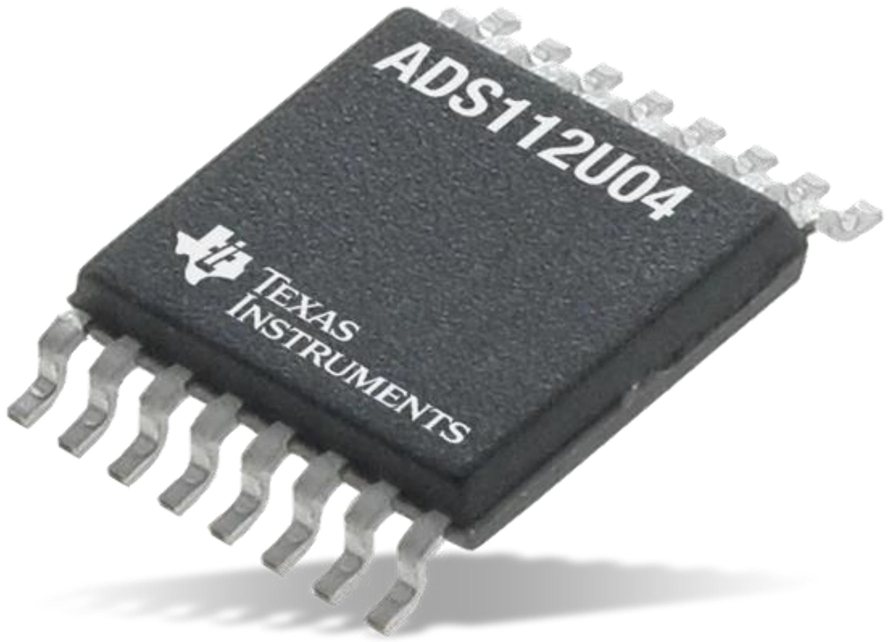




Mixed-Signal ICs

- Combine both **analog and digital** functions.
- Used in complex systems like communication devices and sensors.
- **Examples:**
  - **ADC (Analog-to-Digital Converter)**
  - **DAC (Digital-to-Analog Converter)**
  - **PLL (Phase-Locked Loop)**
  - **Sensor interface ICs**



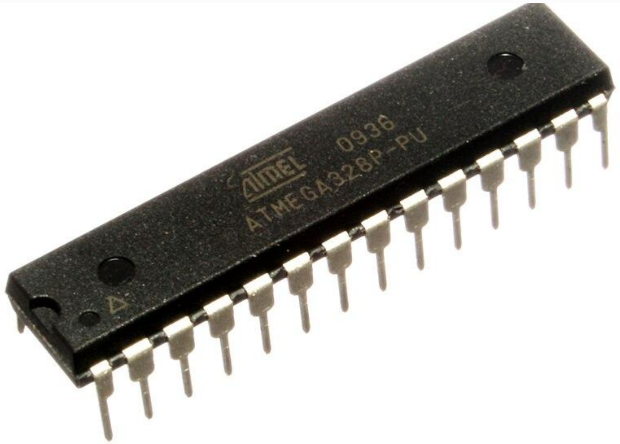


ICs Based on Applications

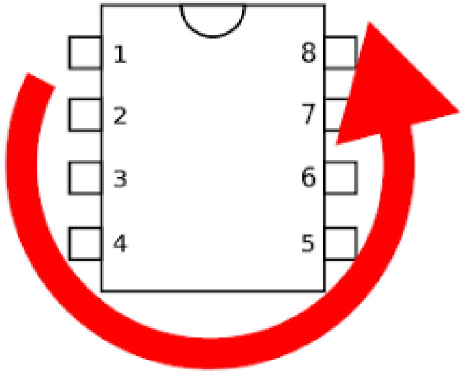
Application Area	IC Example
Computing	CPUs, RAM, ROM
Communication	Modems, RF transceivers
Power	Voltage regulators, power amplifiers
Signal Processing	ADC/DAC, DSP chips
Control Systems	Microcontrollers, PLC ICs
Timing	555 Timer, RTC ICs

IC Packaging

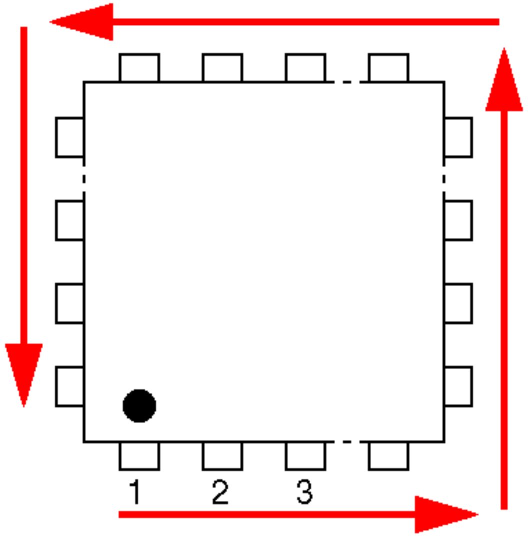
Package	Mounting	Pins Location	Use Case
DIP	Through-hole	2 sides	Prototyping, beginners
SOP	Surface mount	2 sides	Consumer electronics
QFP	Surface mount	4 sides	Microcontrollers, DSPs
BGA	Surface mount	Underneath	High-speed processors
TO	Through-hole	Leads extend	Power devices, regulators
CSP	Surface mount	Underneath	Miniature, high-density ICs



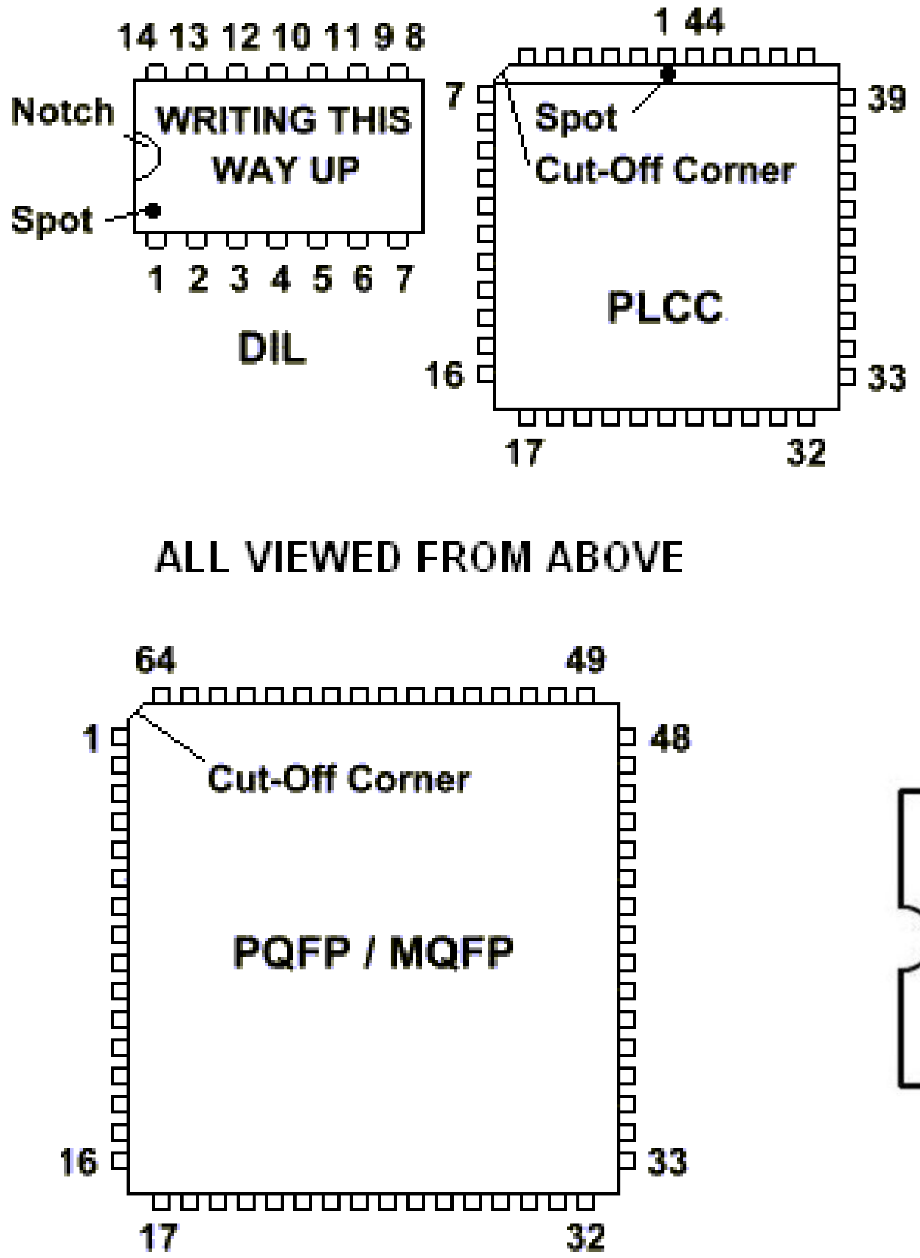
IC Pinout



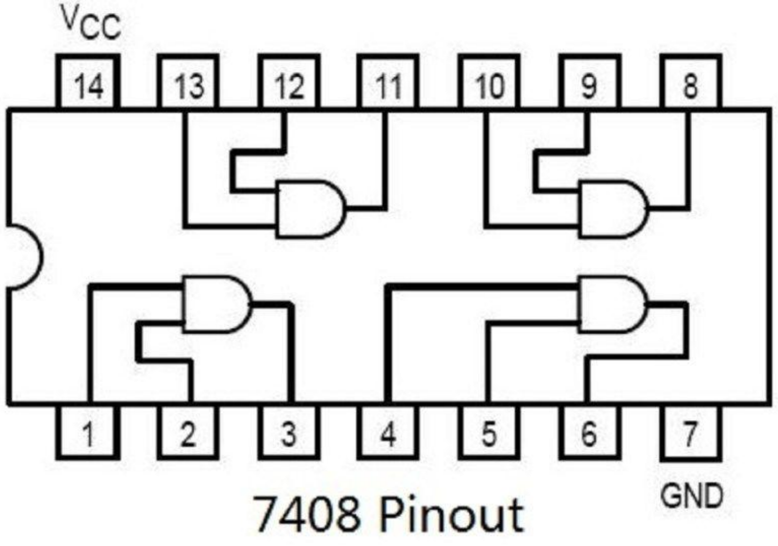
1
2
3
4
8
7
6
5



1
2
3

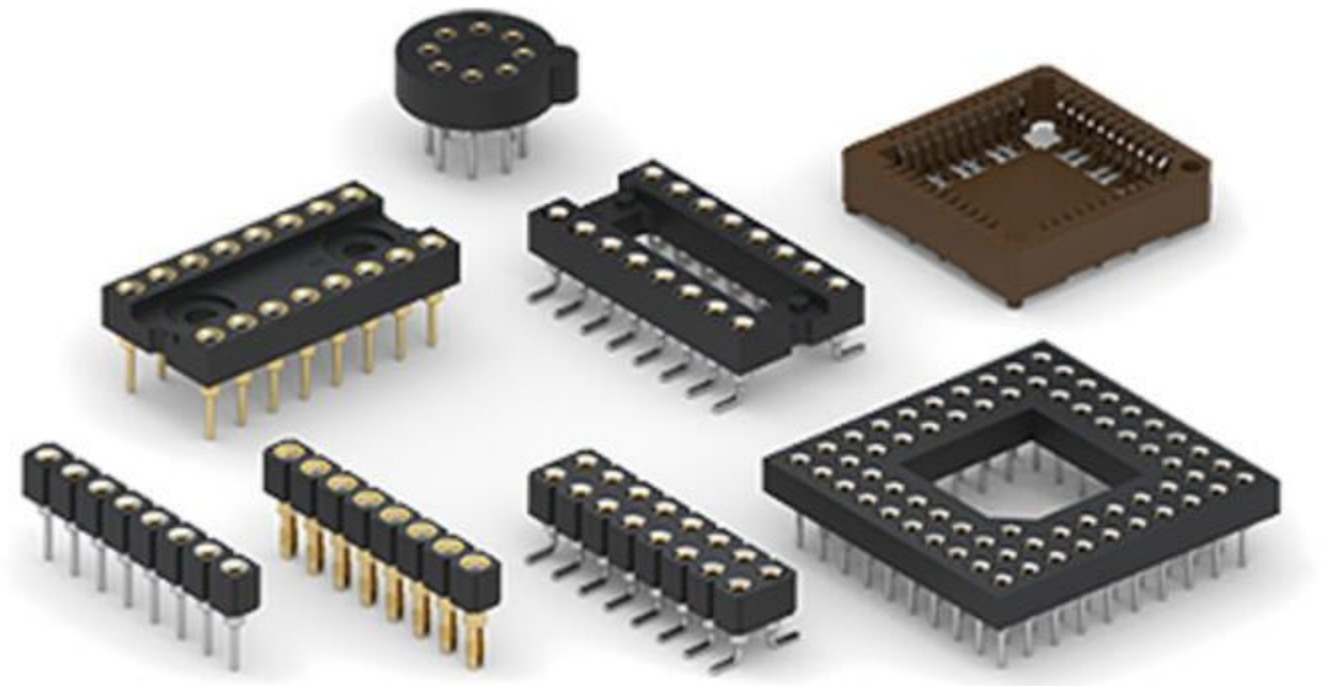


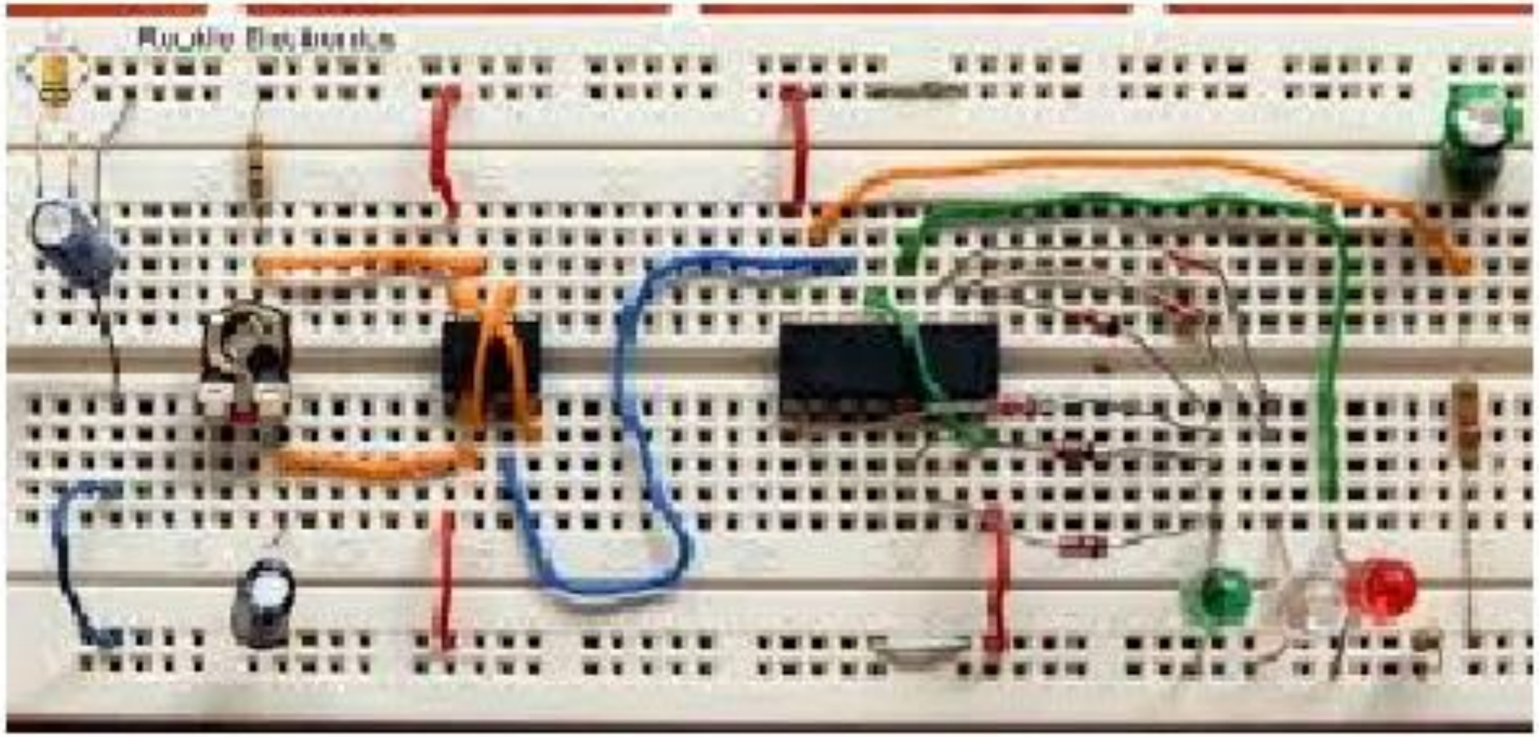
14 13 12 10 11 9 8
Notch
WRITING THIS WAY UP
Spot
1 2 3 4 5 6 7
DIL
1 44
7
39
Spot
Cut-Off Corner
PLCC
16
33
17
32
ALL VIEWED FROM ABOVE
64
49
1
48
Cut-Off Corner
PQFP / MQFP
16
33
17
32



V_{CC}
14
13
12
11
10
9
8
1
2
3
4
5
6
7
GND
7408 Pinout

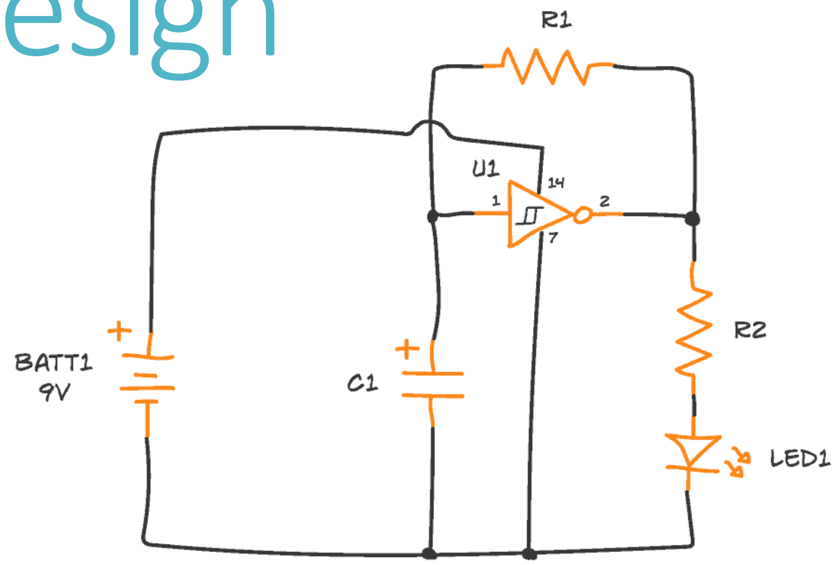
IC Sockets/Bases





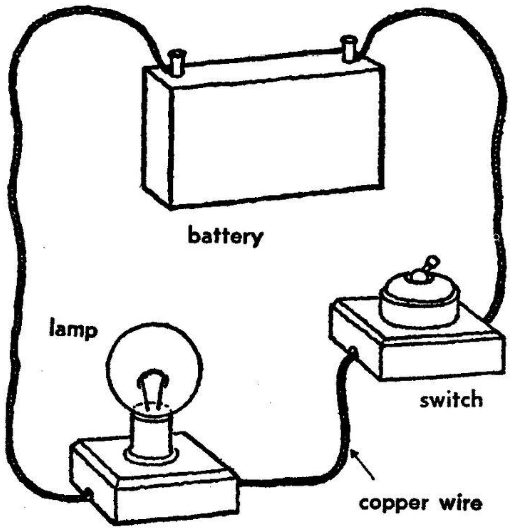

C1
C5
R6
C4
PULSE
T1
R3
IC2
J1
VCC
EN
VOUT
GND
R4 R11 R10 R5 R9
Place finger
SENSOR
R1 R2 R7 R8
Easy Pulse
www.Embedded-Lab.com

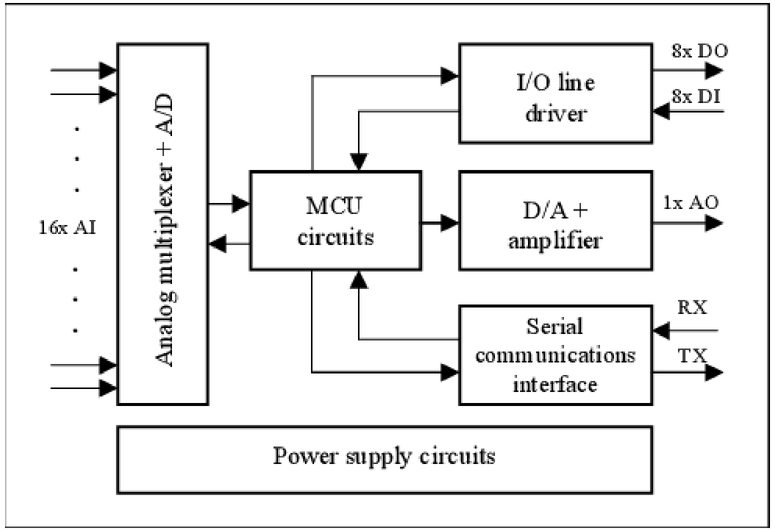
Stages of Electronic Circuit Design



Conceptual Design:

- Define the **goal** of the circuit (e.g., amplifier, sensor interface, power supply).
- Select components: resistors, capacitors, ICs, transistors, microcontrollers, etc.
- Create **block (schematic, wiring) diagrams** to organize subsystems.



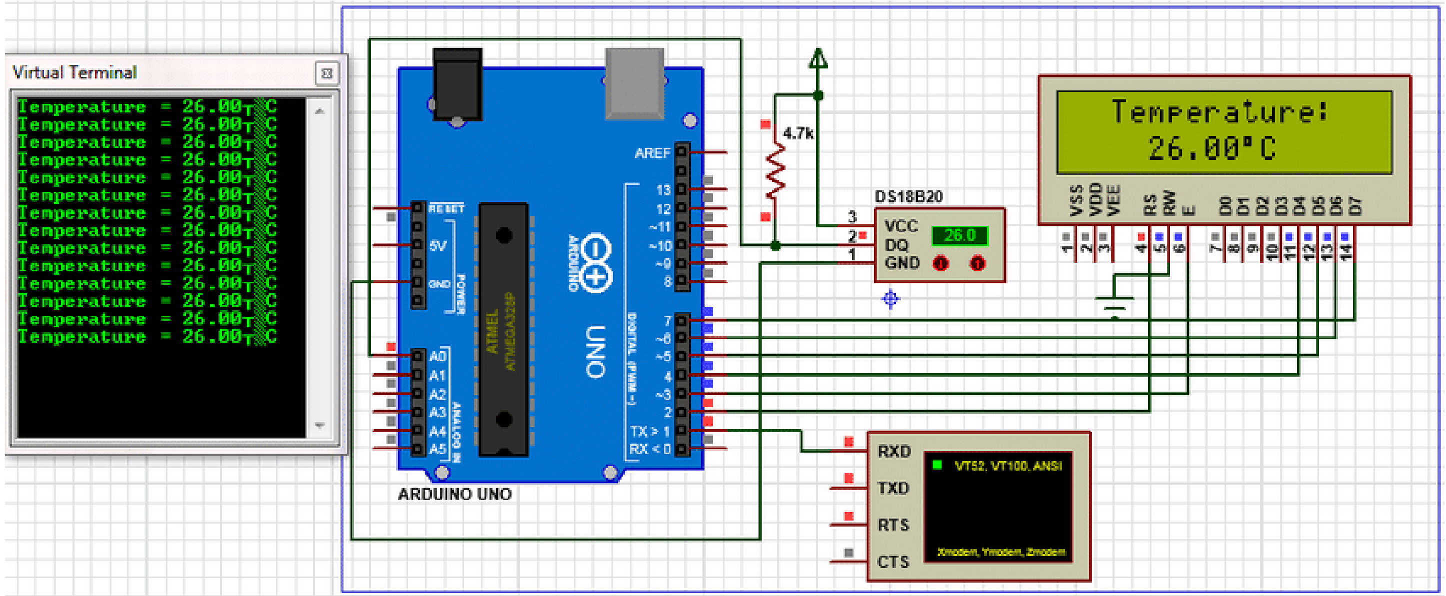


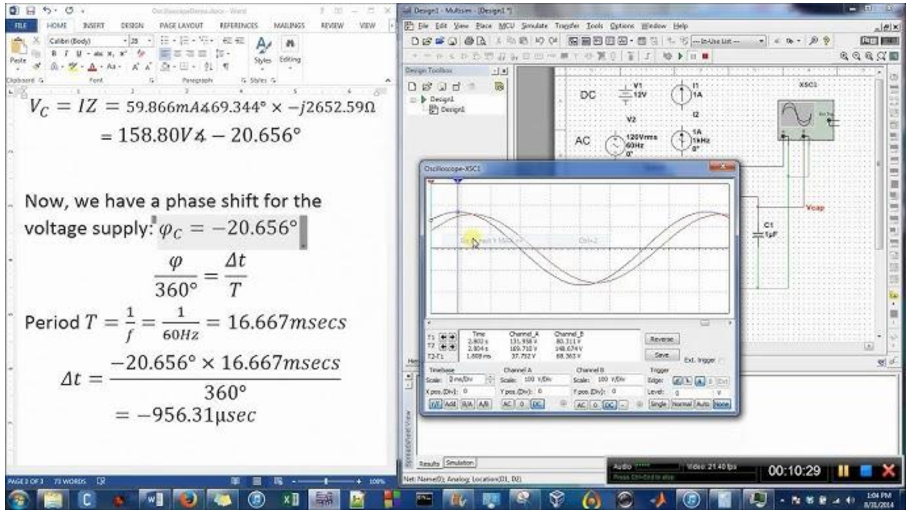
Stages of Electronic Circuit Design Cont.

Circuit Simulation:

Use software (e.g., **LTspice**, **Proteus**, **Multisim**) to simulate the behavior before physically building it.

Verify voltage levels, current, signal timing, frequency response, etc.

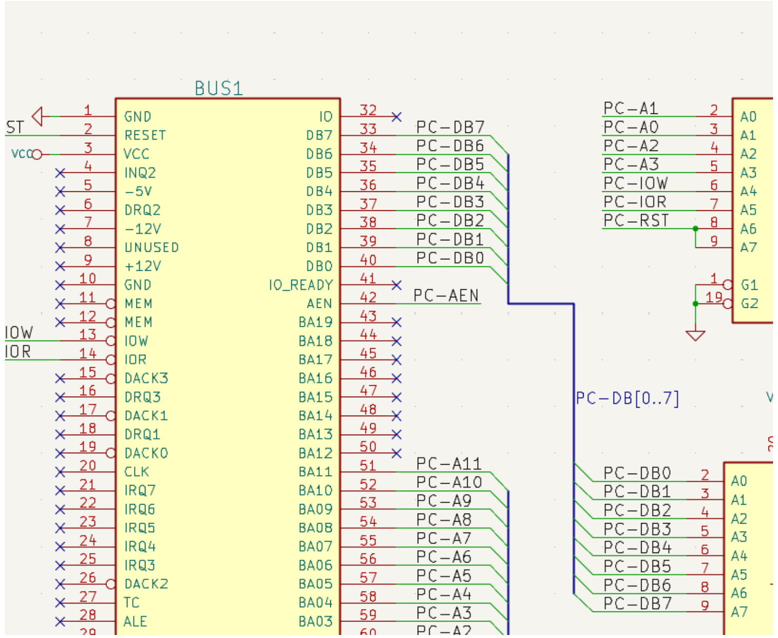


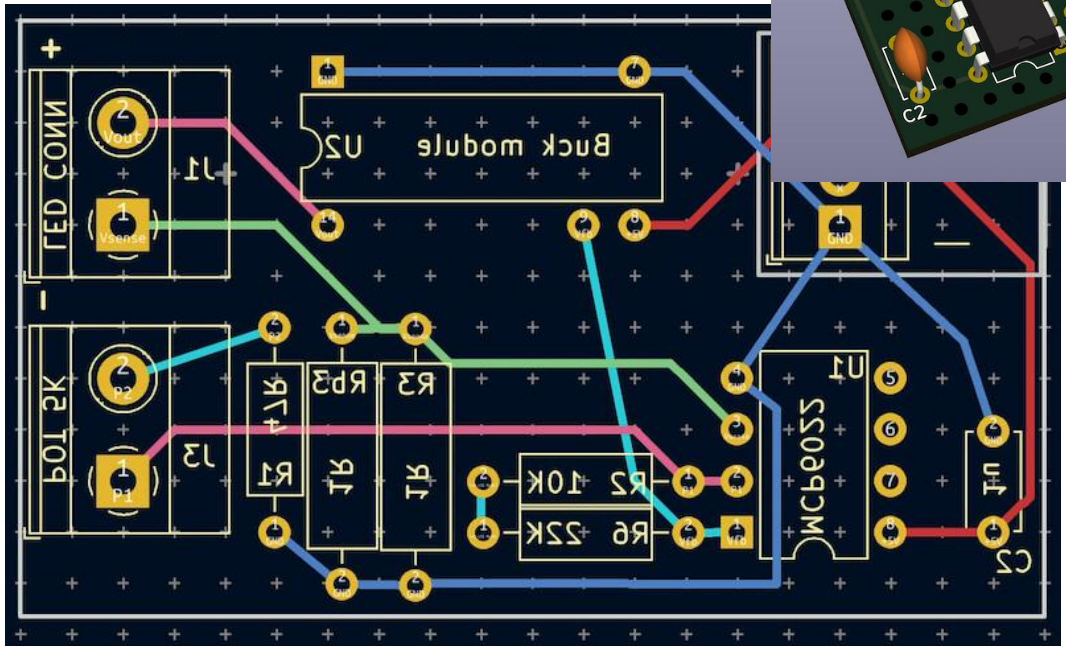


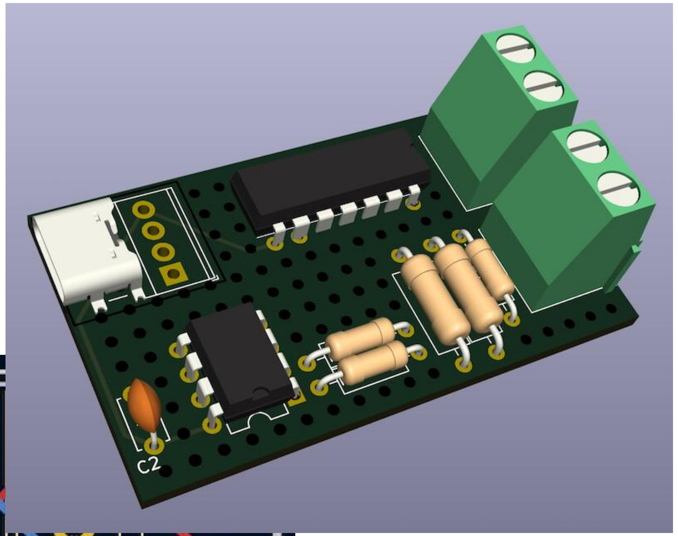
Stages of Electronic Circuit Design Cont.

Schematic/PCB Design:

Use CAD tools (e.g., **KiCad**, **Eagle**, **Altium Designer**) to draw the detailed circuit diagram showing how components are connected.

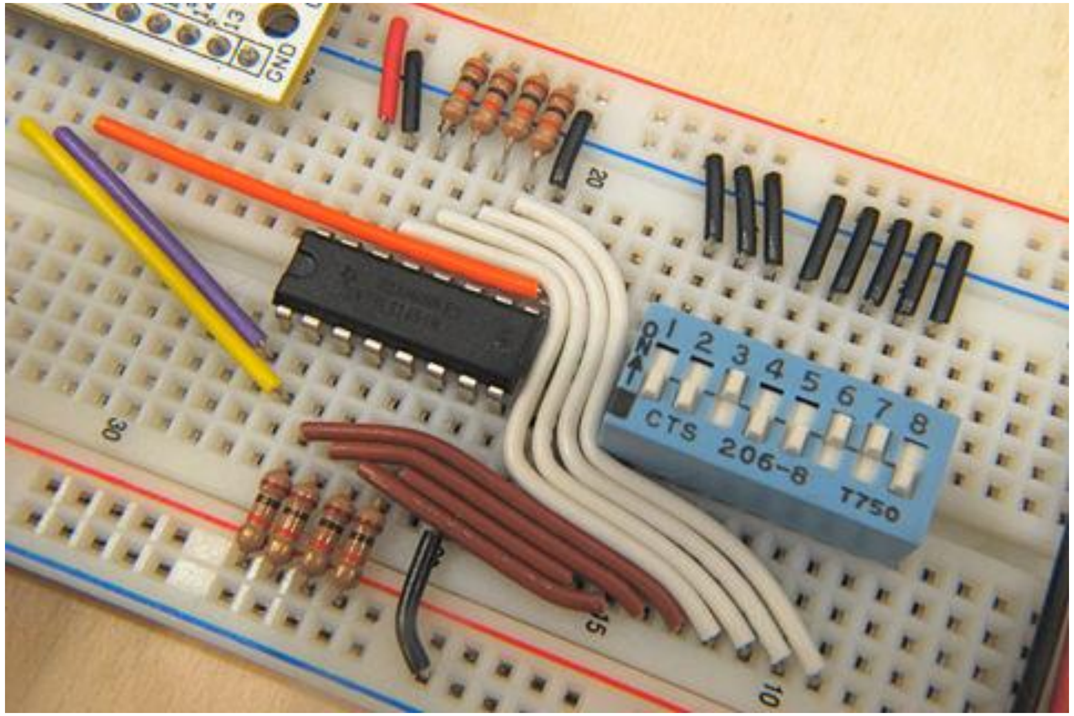






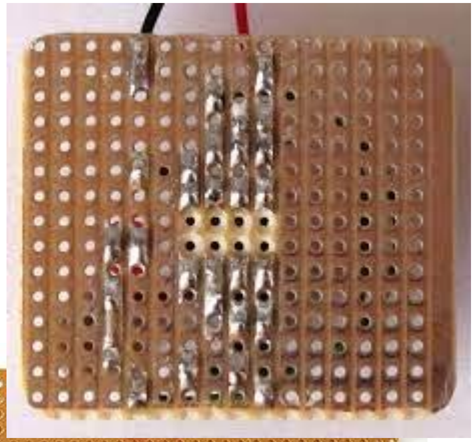
Development and Prototyping Methods: Breadboard

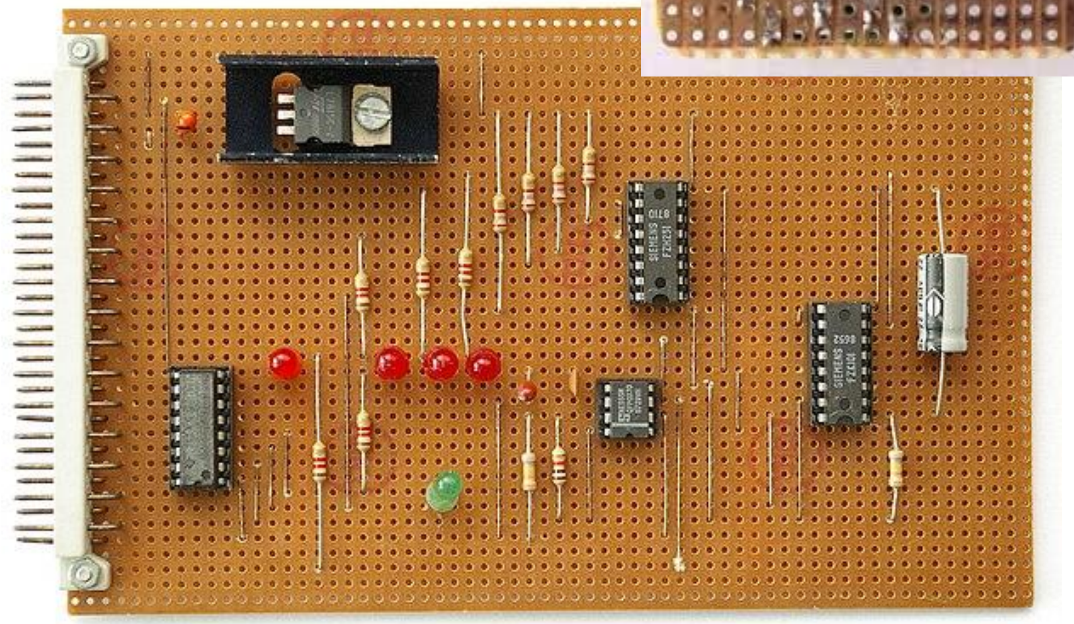
- **What it is**: A reusable platform for building temporary prototypes without soldering.
- **Best for**: Beginners, quick testing, debugging.
- **Advantages**:
  - Easy to change components/wiring.
  - No soldering required.
- **Disadvantages**:
  - Poor high-frequency performance.
  - Prone to loose connections.



Development and Prototyping Methods: Veroboard (Stripboard, Dotboard)

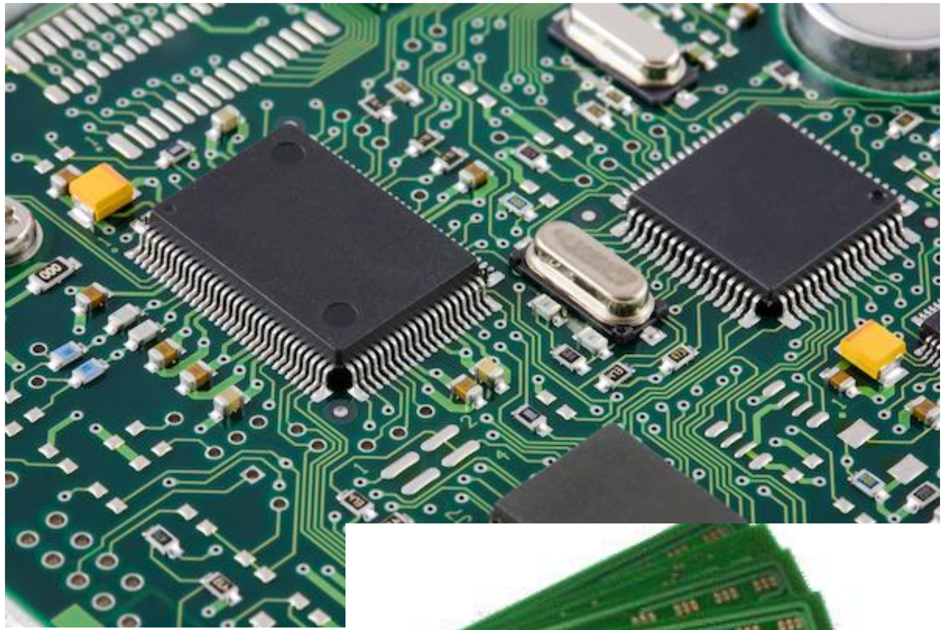
- **What it is**: A board with pre-drilled holes and copper strips for semi-permanent prototyping.
- **Best for**: Intermediate-level projects and small production runs.
- **Advantages**:
  - More robust than breadboards.
  - Easy to solder.
- **Disadvantages**:
  - Limited layout flexibility.
  - Can get messy with many connections.

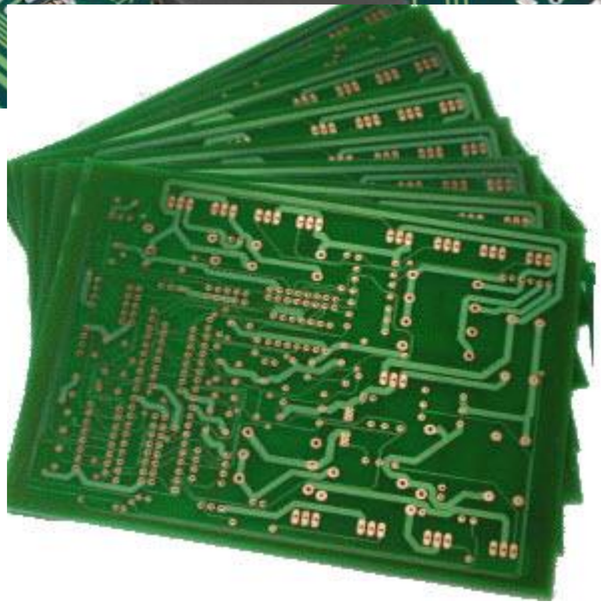




Printed Circuit Board (PCB)

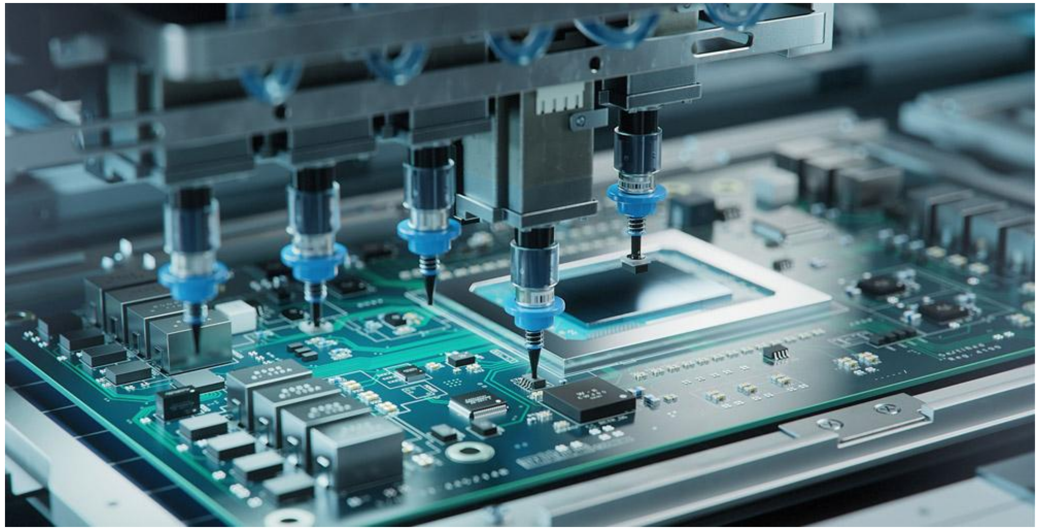
- **What it is**: A board that has copper tracks etched to connect components permanently.
- **Used in**: Commercial and professional electronics.
- **Advantages**:
  - Reliable, compact, and durable.
  - Suitable for mass production.
  - Cleaner signal paths, especially for analog and high-frequency designs.
- **Disadvantages**:
  - Longer development time.
  - Requires design and fabrication tools or services.





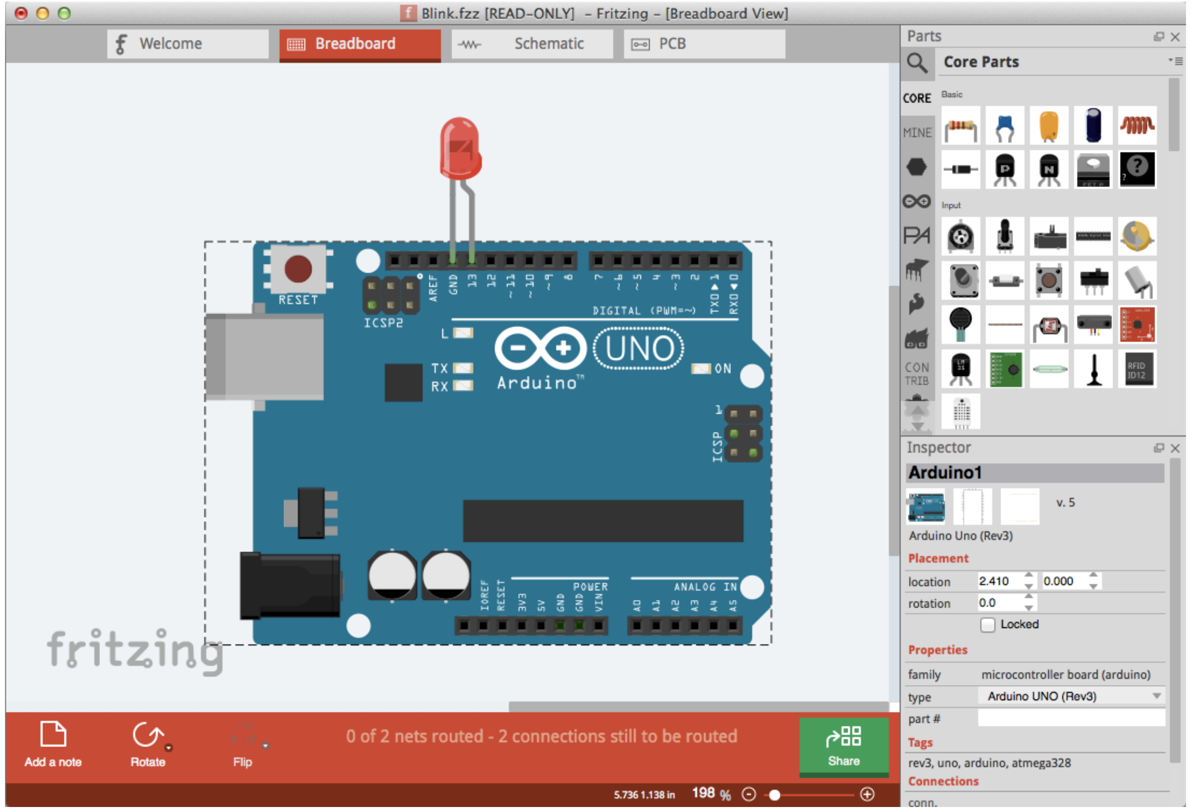
PCB Design Process

1. **Schematic Capture** – define connections using CAD tools.
2. **PCB Layout** – arrange components, define routing on layers.
3. **DRC & ERC** – Design Rule Checks and Electrical Rule Checks.
4. **Gerber File Generation** – standard format sent to PCB manufacturers.
5. **Fabrication and Assembly** – either done in-house or outsourced.



CAD Tools for Circuit Design

Tool	Purpose	Features
KiCad	Schematic + PCB Design	Free, open-source, multi-layer support
Eagle	Schematic + PCB Design	Hobbyist-friendly, good library support
Altium	Professional PCB Design	Advanced features, expensive
Proteus	Simulation + PCB Design	Real-time simulation, microcontroller support
Multisim	Circuit Simulation	Education-focused, SPICE-based
LTspice	Analog Circuit Simulation	Free, industry standard
Fritzing	Beginner-friendly	Breadboard-style schematic + PCB layout



Do-It-Yourself (DIY) PCB Process

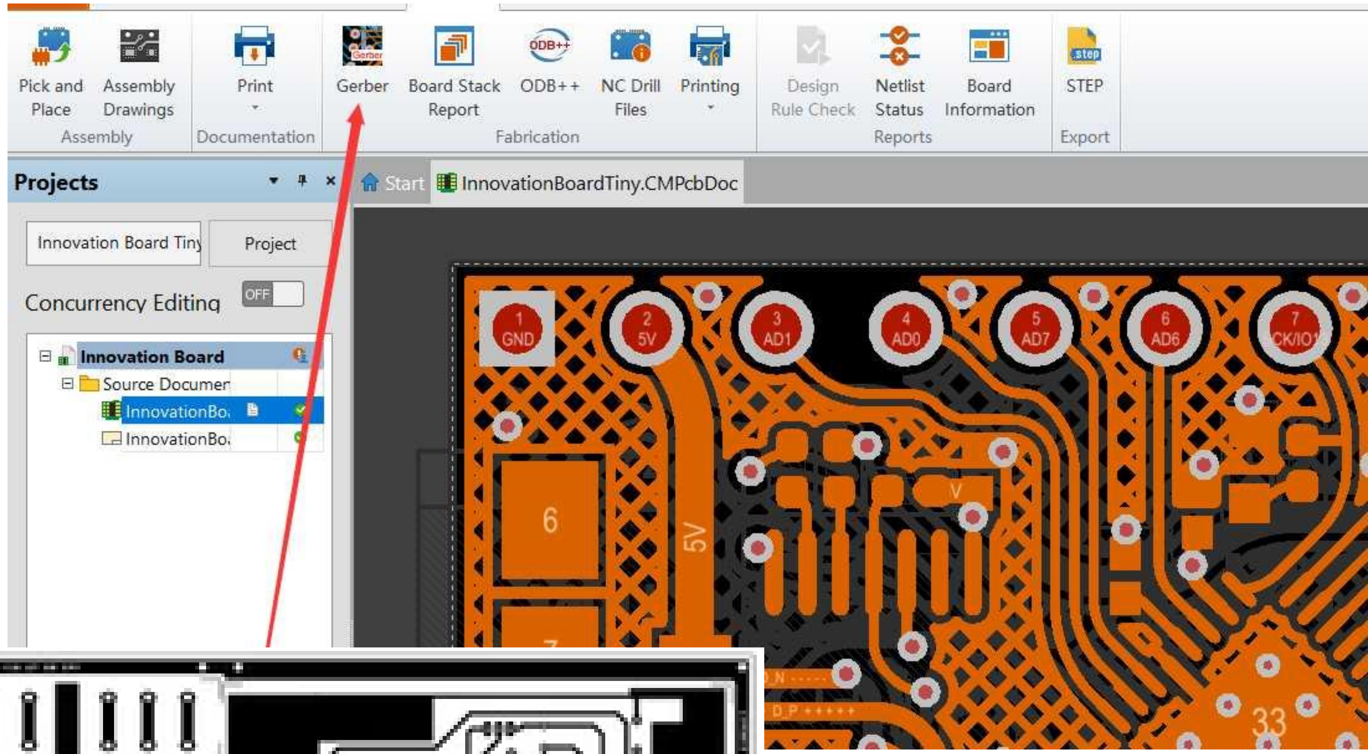
Design the PCB

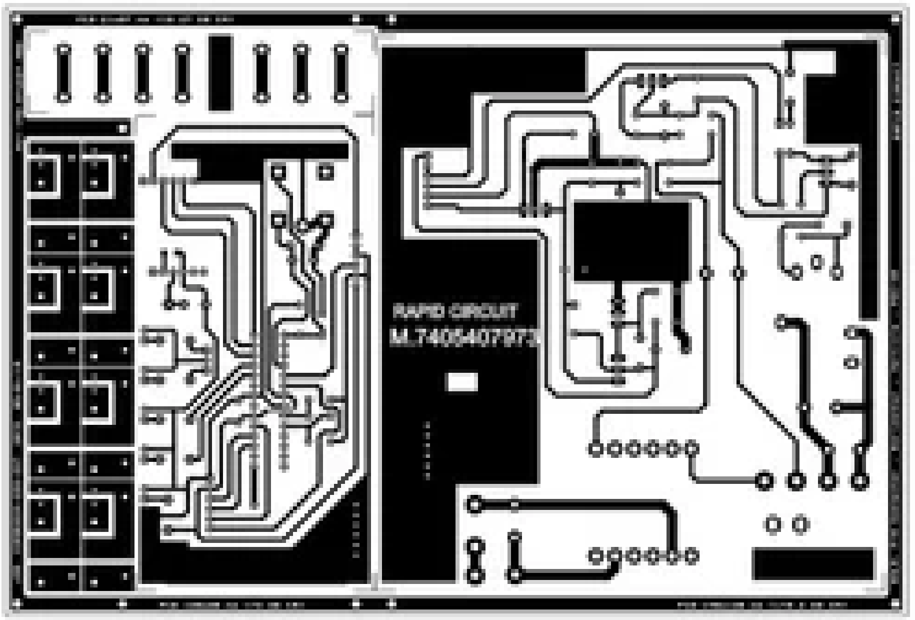
Use CAD tools like **KiCad**, **Eagle**, or **Fritzing**.

Export **Gerber files** (standard file format for PCB fabrication).

Print the Design

Print the **copper layer layout** (mirrored) on glossy paper using a **laser printer**.

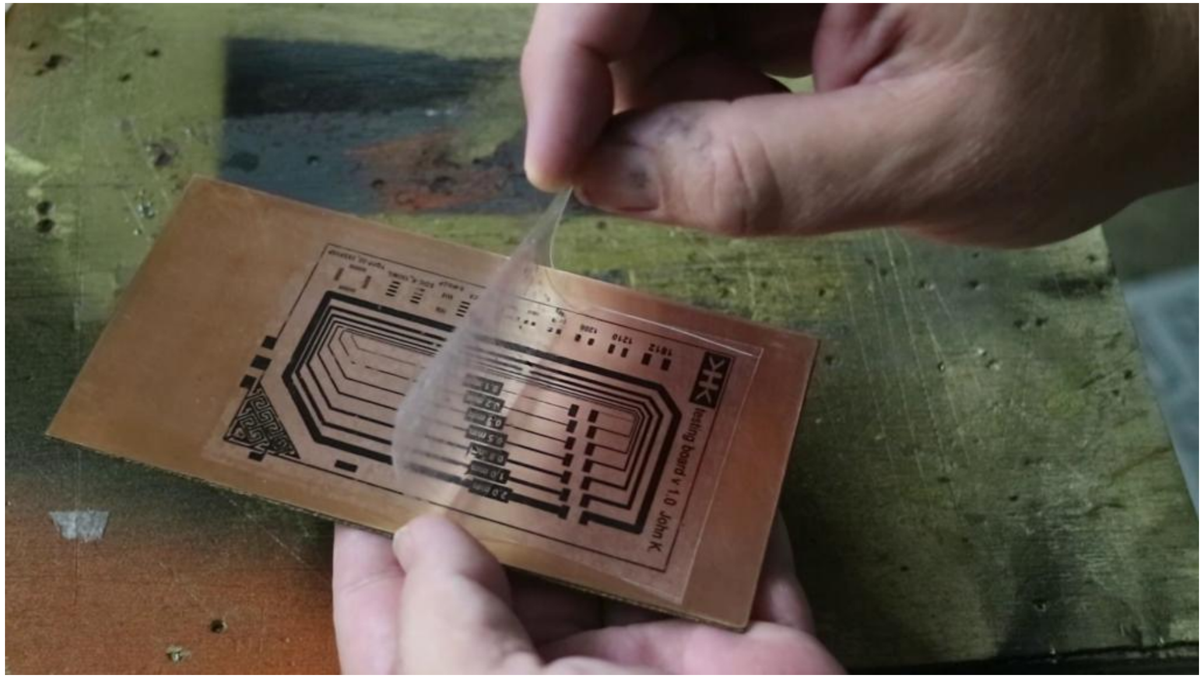




Do-It-Yourself (DIY) PCB Process Cont.

Transfer the Design to Copper-Clad Board

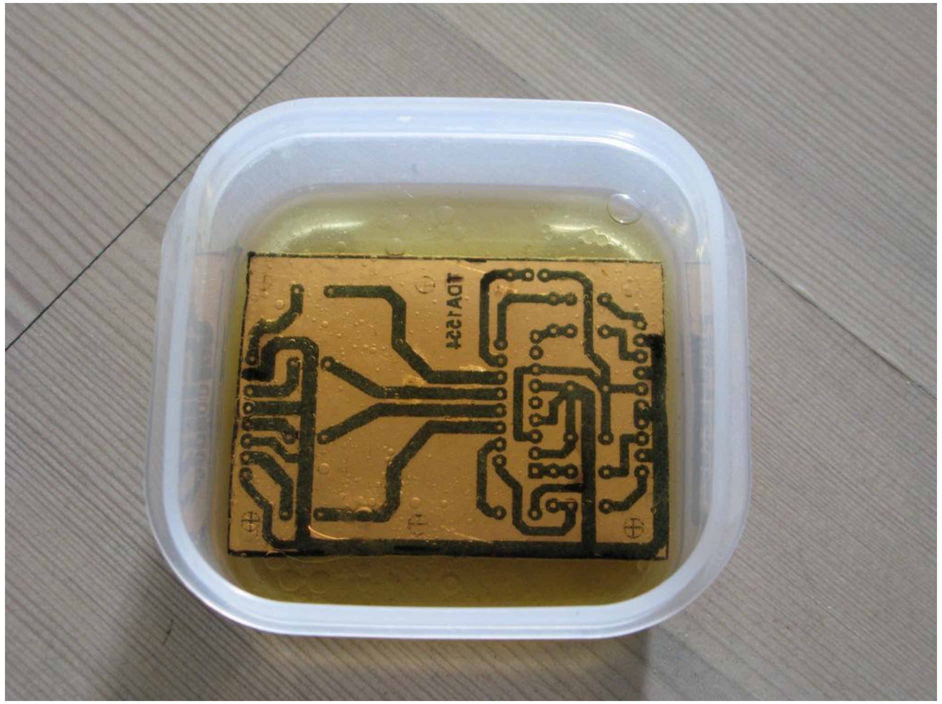
- Place the printed design face-down on a **copper-clad board**.
- Use a **hot iron** or **laminator** to transfer the toner to the copper.

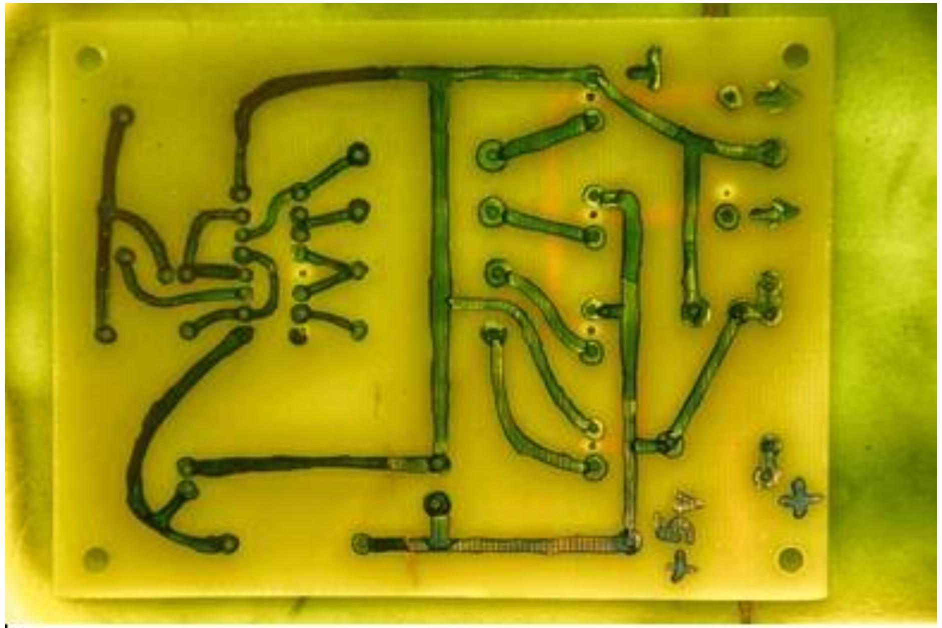


Do-It-Yourself (DIY) PCB Process Cont.

Etching

- Immerse the board in an **etching solution** (like ferric chloride or sodium persulfate).
- The exposed copper dissolves, leaving the PCB traces.





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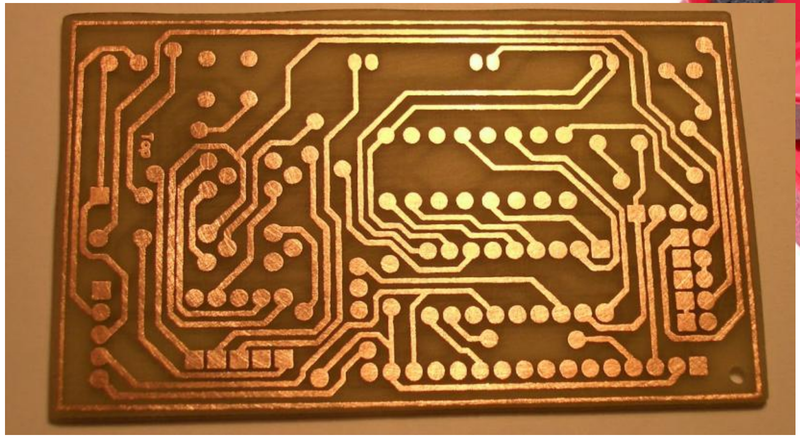
Do-It-Yourself (DIY) PCB Process Cont.



Drilling

Use a **PCB drill or Dremel tool** to make holes for through-hole components.

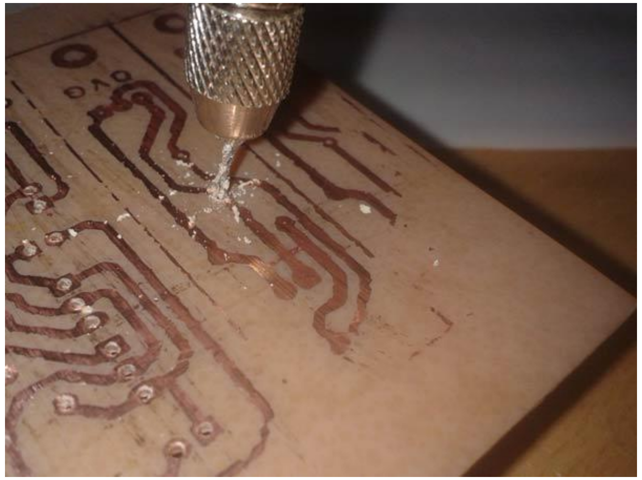
Cleaning and Tinning (Optional)

Remove the toner using acetone.

Optionally, apply solder mask or tin the copper to prevent oxidation.

Component Assembly

Solder components manually.



Thanks